FRC 2024 CRESCENDO -- Robot Design Deep Dive

Design reference for CRESCENDO. Each profile comes from the team's own public material: CAD (Onshape, GrabCAD, Fusion 360, STEP), technical binders, code, and Chief Delphi release and reveal threads. Performance data comes from The Blue Alliance.

The CRESCENDO design landscape

The game

CRESCENDO is a NOTE game. Robots collect NOTES (foam rings) from the floor or the SOURCE. Robots score NOTES in the AMP or shoot them into the SPEAKER. In the endgame, robots climb the CHAIN on the STAGE and can score a NOTE in the TRAP. Many designs used a speaker shooter with a fixed or variable angle, fed by an under-the-bumper intake. Many top teams did not build a TRAP mechanism.

How to use this document

The first section lists the profiled robots in the top 30 by Local EPA. The second section lists the other profiled robots. Each profile gives the archetype, the subsystems, vision, endgame, autonomous routines, the public CAD or binder, and lessons for students. A star marks a study pick. Each study pick has downloadable CAD or a full technical binder and one specific design lesson. "Unconfirmed" means the public material does not state the detail. It does not mean the robot does not have it.

About the EPA numbers

The Statbotics data API is offline. EPA is Local EPA (approx): it was computed from The Blue Alliance match data with the open-source Statbotics model (github.com/avgupta456/statbotics). The rank order follows Statbotics. The absolute values can differ from statbotics.io. Records are win-loss-tie totals from official events on The Blue Alliance.

Profiled robots at a glance (sorted by Local EPA)

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**3005 RoboChargers**	Surge	**54**	#4	59-10	'Surge': double-sided under-bumper intake; 4x NEO Vortex pivoting laun	CD thread
	4414 HighTide	TIDEPOD	**54**	#3	47-10	'TIDEPOD': 4-Kraken shooter on 4.5 in wheels; half-moon wheel sends no	CD thread
■	**1690 Orbit**	Doppler	**52**	#5	66-5-1	'Doppler': low-profile robot; pivot driven by machined pin-tooth pinio	Onshape CAD
■	**2056 OP Robotics**	LOW-KEY	**52**	#6	67-3	'LOW-KEY': shooter on a low-backlash arm pivot; Limelight 3G 2D aiming	Tech binder
■	**604 Quixilver**	Domotron	**49**	#10	58-10-1	'Domotron': shooter and elevator; chain-deployed intake with flexible	Onshape CAD
■	**1778 Chill Out**	Zappy Feet	**49**	#11	43-22	'Zappy Feet': shooter with 2 bottom rollers and 1 top; amp bar on a vi	Onshape CAD
■	**1706 Ratchet Rockers**	Riot	**48**	#16	44-4	'Riot': swerve; belt-driven elevator carrying the amp/trap mechanism;	Onshape CAD
	2075 Enigma Robotics	Masquerade	**48**	#15	46-16-2	'Masquerade': note shooter with a shuttling shot mode; intake deploys	CD thread
	4481 Team Rembrandts	CODA	**48**	#14	39-10	'CODA': REV MAXSwerve; two-sided under-bumper intake; one shooter/amp	CD thread
	368 Team Kika Mana		**46**	#23	40-5	Speaker shooter with split wheels, hook climbers and a trap mechanism;	CD thread
■	**2910 Jack in the Bot**	Typhoon	**46**	#22	50-16-1	'Typhoon': +/- 360 deg turret shooter; two-entry feed loads notes at a	Onshape CAD
	33 Killer Bees	Buzz XXIX	**45**	#30	49-17-2	'Buzz XXIX': 4x Kraken X60 shooter with independent wheel sets, full-w	CD thread
■	**3467 Windham Windup**	Nocturne	**45**	#28	59-22-1	'Nocturne': arm and shooter, harmony climb hooks, leaf-blower trap ass	Onshape CAD
■	**1114 Simbotics**	Simbot Skyfall	**45**	#32	35-28	'Simbot Skyfall' 2.0: rebuilt shooter that amps out the rear via a rol	Onshape CAD
■	**3061 Huskie Robotics**	Artemis	**42**	#66	32-16-2	'Artemis': swerve, intake with IR note sensing, kicker into a pivoting	Onshape CAD
■	**581 Blazing Bulldogs**	Titan	**40**	#78	46-16	'Titan': MK4i swerve; pivoting split-flywheel shooter; angled amp/trap	Onshape CAD
■	**4795 Eastbots**	Perseus	**39**	#93	47-13	'Perseus': 22 lb pivoting shooter/amp tower on 2x NEO Vortex at 144:1	Onshape CAD
■	**6329 The Bucks' Wrath**	Dial	**39**	#95	53-19-1	'Dial': high-pivot shooter that also scores amp and climbs; full-width	Onshape CAD
■	**1787 The Flying Circuits**	Ronnie	**38**	#123	30-11	'Ronnie': pivoting flywheel 'accelerator' (1 Kraken per side); 'snake	Onshape CAD
■	**5951 Makers Assemble**	Off-season robot (name not stated); competition robot: MAroon	**38**	#113	43-21-2	Off-season 2056-concept robot (CAD): under-bumper intake, pivot arm, f	Drive folder / doc
■	**5419 Berkelium**	Radium the Roomba	**37**	#132	26-14-1	'Radium the Roomba': swerve; 4-side ground pickup via street sweepers;	Onshape CAD
■	**131 CHAOS**	Double Time	**36**	#144	45-25-1	'Double Time': under-bumper intake; lift with chain-pull notches for t	Onshape CAD
■	**3357 COMETS**	Vaisala	**36**	#150	47-15	'Vaisala': 24x28 in, 75 lb swerve robot; roller-pinion sector gears; s	GrabCAD
	4043 NerdHerd	Notezart	**36**	#145	46-16	'Notezart': first-year swerve; RI3D-style arm with flywheels and kick-	Onshape CAD
	5406 Celt-X		**36**	#159	41-21	SDS Mk4i swerve; flywheel shooter set by arm angle; roller intake; amp	Onshape CAD
■	**6036 Peninsula Robotics**	Snoopy	**36**	#162	43-19-1	'Snoopy': turreted speaker shooter on an X-contact bearing with auto-a	Onshape CAD
■	**111 WildStang**		**35**	#180	22-23	Swerve; tilting shooter with note compression; lift on the amp note pa	Onshape CAD
■	**2877 LigerBots**		**35**	#192	35-27-1	Under-bumper intake; shooter pivots on an angled cascade elevator; win	Onshape CAD

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**9312 NERD Spark**	BATMAN	**35**	#187	33-23	'BATMAN': swerve; intake-shooter tilted by a printed CF 4-bar; 2 DOF a	Onshape CAD
■	**8033 Highlander Robotics**	Banshee	**34**	#208	33-13	'Banshee': pivoting shooter on a Maxspline shaft, 2 independent wheel	Onshape CAD
■	**341 Miss Daisy**	Miss Daisy XXII	**33**	#247	38-22-1	'Miss Daisy XXII': full-width over-bumper intake feeding an under-bump	Onshape CAD
■	**2930 Sonic Squirrels**	Maestro	**33**	#238	29-26-1	'Maestro': continuous belt-driven elevator; PhotonVision note detectio	Onshape CAD
■	**930 Mukwonago BEARs**		**32**	#290	35-16-1	Turreted shooter driven through a strain-wave gearbox; 3D printed gear	Onshape CAD
■	**2102 Team Paradox**		**32**	#262	19-10	Combined intake/shooter with two roller pairs (shoots from either side	Onshape CAD
	7312 T3		**32**	#293	37-22	Fixed-angle two-sided shooter with rotary amp bar, OTB rotary intake,	Onshape CAD
	1710 The Ravonics Revolution	ADONIS	**31**	#316	29-13	'ADONIS': swerve robot with an intake, wrist, shooter and a rope-rigge	Fusion 360 CAD
■	**1155 SciBorgs**	Taiwanese Fried Chicken	**30**	#334	18-9	'Taiwanese Fried Chicken': pivoting flywheel shooter that also scores	Onshape CAD
	3136 O.R.C.A.	JT	**30**	#367	38-23-1	Arm-mounted intake that shoots notes out its back with two flywheels;	Onshape CAD
■	**3729 The Raiders**	Mr. Mimlitz	**30**	#328	15-13-1	'Mr. Mimlitz': swerve; hybrid under-bumper intake with slapdown and pr	Onshape CAD
	5431 Titan Robotics	Orpheus	**30**	#341	22-23	'Orpheus': swerve; V2 rack-and-pinion shooter angler (about 35-55 deg)	Fusion 360 CAD
■	**1414 IHOT**	Encore	**29**	#420	41-21	'Encore': Orbit-inspired layout; differential speed shooter; amp defle	Onshape CAD
■	**6647 PrepaTec - VOLTEC**		**29**	#415	34-21-1	Swerve NOTE robot; elevator removed mid-season; Kraken shooter and int	Onshape CAD
■	**2491 NoMythic**	Eurydice	**27**	#475	17-8	'Eurydice': 26x26 in swerve; bottom-pivot two-roller shooter that also	Onshape CAD
■	**7525 Pioneers**		**27**	#523	22-13-1	Static-shooter swerve robot (SDS MK4i); beater-bar intake; roller amp	Onshape CAD
	1038 Lakota Robotics	Vio-Linda	**25**	#663	14-16	'Vio-Linda': REV MAXSwerve; under-bumper intake; 'TRAG' amp/trap score	Onshape CAD
	3749 Team Optix		**25**	#622	17-9	Single-jointed arm carrying a four-bar intake that also shoots; two Li	Onshape CAD
■	**6838 X-SHARC**	Blackout	**25**	#660	15-11	'Blackout': swerve; shooter on a pivoting arm; touch-and-grab ground i	Onshape CAD
	2423 The KwarQs	ConDUCKtor	**24**	#731	20-23	'ConDUCKtor': over-the-bumper intake; one pivot for shooter and climbe	Onshape CAD
■	**5987 Galaxia in memory of David Zohar**	Andromeda	**24**	#693	30-26	Note robot on flipped swerve; polycarbonate spring-hook Stage climber	Onshape CAD
■	**3737 Roto Raptors**	Synthwave	**22**	#850	25-18-2	'Synthwave': 3D-printed swerve; under-bumper intake; through-robot not	Drive folder / doc
■	**6925 W.A.Robotics**	VENDETTA	**22**	#878	26-21	'VENDETTA': fixed-angle 2-Kraken shooter; over-the-bumper intake with	Onshape CAD
■	**4276 Surf City Vikings**		**19**	#1150	15-9-1	Intake/shooter on a NEO-driven 4-bar, rebuilt mid-season; floor intake	Onshape CAD
■	**663 Power Sharks**	Hammerhead	**17**	#1418	18-23	'Hammerhead': Ri3D-based wide intake-shooter; sliding 8020 extending c	Onshape CAD
	1072 Harker Robotics		**16**	#1561	11-19	Two-stage cascade elevator carrying a pivoting top-down shooter; over-	Onshape CAD
■	**1746 Team OTTO**		**16**	#1591	30-17	CTRE swerve; Falcon shooter fed by a primer; intake rebuilt on a Falco	Onshape CAD
	9047 SESI SC TECHMAKER	Seu Wilson	**15**	#1620	7-3-1	Low robot: fixed under-bumper intake-shooter; elevator that climbs and	Onshape CAD
■	**3596 Rocket Robotics**	Kerchunk	**11**	#2329	3-7	'Kerchunk': Kitbot launcher on a 27x27 in swerve chassis; under-bumper	Onshape CAD

■ = study pick: downloadable CAD or a full technical binder, and one specific design lesson. EPA is Local EPA (approx), computed from The Blue Alliance match data. Records are win-loss-tie totals from official events.

Top-30 robots by Local EPA

■ **Study pick** **3005 RoboChargers** Surge — USA-TX — **Local EPA 54 (#4)** — 59-10

'Surge': double-sided under-bumper intake; 4x NEO Vortex pivoting launcher with amp diverter; telescoping climber; swerve

SHOOTER (SPEAKER) Pivoting launcher, per the reveal post. The team states it uses four NEO Vortex motors for mechanical simplicity: there is no mechanical linkage between the four corners of the launcher. A speed difference between the two sides puts spin on the note. Surgical tubing counterbalances the launcher.	**INTAKE** Under-the-bumper, double-sided intake, per the reveal post. A team member states the intake was upgraded several times during the season and moved between the competition robot, the practice robot and a spare intake. Roller and motor details are unconfirmed (see CAD).
FEEDER / AMP / TRAP A diverter on the launcher allows both amp and speaker shots, per the reveal post. A team member mentions a feeder motor that was never removed during the season. Feeder layout and any trap mechanism are unconfirmed (see CAD).	**DRIVETRAIN** Swerve. A team member states the drivetrain runs NEO Vortex motors, and that the team rotates used swerve modules into the spares pool so spares are 'known good'. The same member credits SPARK Flex controllers with fewer module and motor swaps this season. Module make and gear ratio are unconfirmed; a direct question about the module went unanswered in the thread.
ENDGAME Single telescoping climber, per the reveal post. The team states one part on the robot keeps notes from getting stuck on the climber. Trap and harmony capability are unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release)
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** CAD release (CD)

LESSONS FOR STUDENTS

- Block CAD released next to the finished robot: open both Onshape documents to see how the layout model drives each subsystem. The team states that links from the robot to the block CAD cannot be opened by right-click.
- Four independent launcher motors remove the mechanical linkage between the launcher corners. The team states the same spin could be made without four motors, so the choice was for mechanical simplicity.
- SRPP as a polycarbonate replacement: the team lists lower density, higher tensile stiffness, Loctite resistance and laser cutting as the reasons. The team also states it gives little to no accuracy benefit over polycarbonate.
- A spares practice: used swerve modules move into the spares pool so that spares are known to work.
- The block CAD method is documented in a FIRST Mentor Conference talk by a 3005 member, posted 2024-08-30, with video and slides, as a follow-on to the robot CAD.

CD thread: CAD release (CD) · CAD release (CD) · Top-level robot CAD (Onshape) · Block CAD (Onshape) · Reveal 'Surge' (CD) · Reveal video · Block CAD talk (CD) · Block CAD talk video · Block CAD talk slides (Google Slides)

Profile	4414 HighTide TIDEPOD USA-CA	Local EPA 54 (#3) 47-10

'TIDEPOD': 4-Kraken shooter on 4.5 in wheels; half-moon wheel sends notes to an amp elevator

SHOOTER (SPEAKER) Two-sided wheel shooter. Wheels are 4.5 in lexan tube with 3D printed end caps, wrapped in cat's tongue. The team states 1.5 in compression, 5 in center-to-center, and 1:1.25 upduction on 4 Krakens, 2 per side, with each side linked. Double-sided HTD 5 mm belts drive the wheels. There is no side-to-side compression; most shots use about a 5% speed difference between sides.	**INTAKE** Intake design unconfirmed (not stated in the public release). A team post on 2024-05-03 describes a mechanism based on the '254-2022 cube intake' but does not name the subsystem. That mechanism uses laser-cut 7075 plates, roller bearings and delrin, packaged around a 2x1 tube.
FEEDER / AMP / TRAP A belted half-moon wheel in the shooter has three states. It hard-stops notes in the shooter, flips up out of the way to shoot, and spins in reverse to send the note over the top of the shooter wheel into the amp elevator. It is belted 1:1 off a Kraken, so the motor's internal absolute encoder handles position control. The team calls it 'hacked in'; the belt slipped once, in the last match of the season, and hurt shooting.	**DRIVETRAIN** Drivetrain unconfirmed (not stated in the public release).
ENDGAME Climb, trap and harmony unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines unconfirmed (not stated in the public release). The team won the Autonomous Award at the Ventura County Regional.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Robot reveal (CD)

LESSONS FOR STUDENTS

- A shooter specified in one post: wheel diameter and material, 1.5 in compression, 5 in center-to-center, a 1:1.25 ratio and 4 motors.
- Retest a borrowed trick on the new geometry: reversing the shooter wheels lifted notes on the team's turret robot but not with 4.5 in wheels, so the team added a half-moon wheel.
- A mechanism belted 1:1 to its motor can use the motor's internal encoder for position control, as the half-moon does. A slipped belt then costs accuracy, as it did in the team's last match.
- Tension belts by reprinting: idlers are toothless pulleys on shoulder bolts, and the team reprints an idler 2-3 times after each spacing or ratio change.
- Match parts to build speed: the team kept a traditional bearing stack instead of a large X-contact bearing because it builds the robot in under a week after design lock.

CD thread: Robot reveal (CD) · Robot reveal (CD) · Reveal video · Robot teaser (CD)

■ Study pick

1690 Orbit Doppler

Israel

Local EPA 52 (#5)

66-5-1

'Doppler': low-profile robot; pivot driven by machined pin-tooth pinions; amp and trap mechanisms; swerve

SHOOTER (SPEAKER)
Shooter on a pivot. The pivot pinions are machined from two plates with steel pins between them that act as teeth (team reply). The shooter is removable: clamshell plates with interlocking 'Lego'-style edges clamp it to custom blocks on the main shaft; the team says this is more rigid than screws alone. Two mechanical stops below the shooter edges hold it in travel mode, and a Hall effect sensor detects the arm at the bottom of its travel. A team member wrote that other teams' spin shooters worked better than Doppler's. Flywheel motors and ratios unconfirmed (see CAD).

INTAKE
Not described in the release thread. A team member wrote that other teams' larger-diameter intakes worked better than Doppler's. Rollers and motors unconfirmed (see CAD).

FEEDER / AMP / TRAP
The robot has an amp mechanism and a trap mechanism, per a team member's post. In the trap mechanism a servo pulls a thin 3D-printed lever to release the arm. A constant force spring pulls the arm forward, and a torsion spring in its 3D-printed pivot rotates it. The same team member wrote that downward-shooting amp mechanisms, and trap mechanisms that get closer to the trap, worked better than Doppler's. Amp mechanism details unconfirmed (see CAD).

DRIVETRAIN
Swerve. The team uses an MA3 absolute encoder on its swerve modules (team reply). Module type unconfirmed (see CAD).

ENDGAME
The robot has a trap mechanism with a servo release and spring-driven arm (team replies). The robot slows its drive in the climb state. Stage climb hardware and harmony unconfirmed (see CAD).

AUTONOMOUS
The team ran a 7 note auto, per a team post in the reveal thread. Paths come from the team's own Pathfinder tool. It writes a CSV that the robot parses before each match, with velocity and acceleration limits. Note detection lets the robot skip a missing midline note. In the auto run through most of the Championship playoffs, the robot looks for the first midline note; if it is absent, it jumps ahead and drives to the third. Won the Autonomous Award at ISR District Event #1.

VISION
Three cameras, per the team's posts. A Limelight at the back (shooter side) and an Arducam running PhotonVision at the front read AprilTags. A second front camera detects notes. The June 2024 software sessions covered pose tracking and shooting on the move. Note positions come from the camera's Tx and Ty with trigonometry. The team scales its own odometry by a carpet factor measured in practice matches: drive the field end to end and compare.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Pin-tooth pinions: two machined plates with steel pins as teeth drive the pivot; the team says they are less complex to make than they look.
- Read a top team's own critique: a team member wrote that other teams' shooter, intake, amp and trap concepts worked better, and that low height was the one thing they perfected.
- Build a fallback into an auto path: when the camera does not see the first midline note, Doppler's auto jumps to a later path point and goes for the third note.
- Calibrate odometry to the carpet: the team drives the field end to end in a practice match and scales odometry by the measured ratio.
- Integrate CAD often: the team defines subsystem volumes first, gives each subsystem a CAD lead, and integrates about 3 times a day.

Onshape CAD: Onshape CAD · Onshape CAD · Top-level x_t files (Google Drive) · CAD release thread (CD) · Behind the Bumpers video · Reveal thread (CD) · Software session slides (Google Slides)

■ Study pick

2056 OP Robotics LOW-KEY

Canada

Local EPA 52 (#6)

67-3

'LOW-KEY': shooter on a low-backlash arm pivot; Limelight 3G 2D aiming; pneumatic climber; swerve

SHOOTER (SPEAKER) Upper and lower flywheels; the arm sets the shot angle. Changed after the first district event: the team tipped the flywheels downward for a lower release angle, so the robot could score from the wing line. The team states the arm could not go low enough for that shot. The feed position did not change, so the note handoff was not affected.	**INTAKE** Rollers of 1 in OD, 3/8 in ID 40A polyurethane stretched onto live 1/2 in rounded hex shafts. Timing pulleys in the middle of the rollers let the timing belts act as conveyor belts. Side kicker rollers are 2 in OD stealth wheels. One Kraken X60 drives the intake through a 2:1 reduction for a linear speed of 25 ft/s, per the team.
FEEDER / AMP / TRAP Shooter feed rollers are blue 1 in OD sushi rollers on a 1/2 in hex shaft. The arm drives to an amp scoring position with CTRE Motion Magic and tracks the speaker with a PD loop with no gravity term. The arm gearbox is custom, with every gear bonded to its shaft with Loctite 609. Trap mechanism unconfirmed (not stated in the public release).	**DRIVETRAIN** Swerve, per a team post of 2024-10-25 on module wiring. CAN runs from the first motor to the encoder, to the second motor, then to the next CAN device. CAN and power wires pass through a cut-out in the top module cover and are soldered to the CANcoder board. Module model unconfirmed (see binder).
ENDGAME The climber uses pneumatic quick exhaust valves, which the team buys from Automation Direct. Climber geometry, trap and harmony unconfirmed (see binder).	**AUTONOMOUS** In auto, chassis yaw, tx and distance to the tag correct the odometry estimate. Note count and paths unconfirmed (see binder). TBA lists the Autonomous Award at McMaster, Newmarket and the Ontario Science Division. The code uses the WPILib timed framework, not command-based.
VISION Limelight 3G on LimelightOS with no coprocessor. It tracks the center speaker AprilTag in 2D: tx turns the chassis to aim and ty gives distance, which sets the arm angle. The team did not use MegaTag or 3D pose estimation. It states that 2D tracking runs at a higher FPS and needs no filtering or latency compensation.	**PUBLIC CAD / DESIGN** Technical binder (PDF)

LESSONS FOR STUDENTS

- Backlash control without shim tape: Loctite 609 fills the gap in each gear bore, and the final stage uses a broached sprocket and a 1 in long machined gear hub instead of a 1/2 in deep gear bore.
- Change the cheap part first: tipping the flywheels downward lowered the release angle for wing-line shots without a new arm range or a new note handoff.
- Simple vision can be enough: a Limelight 3G tracking one AprilTag in 2D (tx, ty) aimed the chassis and set the arm angle, with no pose filtering or latency compensation.
- Match the controller to the move: CTRE Motion Magic for large arm moves such as the amp position, and a PD loop with no gravity term for tracking a moving setpoint.
- The team states why it builds custom gearboxes instead of planetaries: custom packaging and chain tensioning, in-house manufacturing, and a way to remove backlash in a long arm.

Tech binder: Technical binder (PDF) · Technical binder release thread (CD) · Technical binder (PDF) · CAD drawing package (PDF) · Robot teaser video

■ Study pick

604 Quixilver Domotron

USA-CA

Local EPA 49 (#10)

58-10-1

'Domotron': shooter and elevator; chain-deployed intake with flexible polycarbonate rollers; drivetrain unconfirmed (see CAD)

SHOOTER (SPEAKER) Unconfirmed (see CAD). A team member stated the shooter hex shafts were cut shorter on the robot than the CAD shows.	**INTAKE** Deployable intake with polycarbonate rollers. The team made the rollers flex in an impact; a roller could flex more than half its diameter in a bad hit. Short shaft segments replace a full-width hex shaft, which would bend permanently. The left and right sides are tied together through chain and a cross shaft inside the frame perimeter, so no rigid cross bars are used. The team chose 3/8 in bearings for weight and fit in the tube plugs, and used 3/8 in OD spacers in place of hex. The team reports two failure points it did not solve in season: the bolts holding the axles to the side plates sheared, and the deploy chain snapped.
FEEDER / AMP / TRAP The robot has an elevator. A team member stated two hex shafts on the elevator came from an earlier climber iteration. Feeder, amp and trap mechanisms are unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). A team member stated the bellypan holes were hand-drilled, not CNC-cut, and that the CAD omits most of them. Electronics are suppressed in the CAD to reduce lag; the team planned the layout in advance.
ENDGAME The robot has a climber. The team reused the top elevator hex shaft to tension the climber with surgical tubing. The tubing is not in the CAD.	**AUTONOMOUS** Routines unconfirmed (see code). The team won the Autonomous Award in the Milstein Division.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Design the intake to flex instead of resisting impact: polycarbonate rollers on short shaft segments can deflect more than half their diameter without bending a shaft.
- Tie the two sides of a wide intake together through a cross shaft and chain inside the frame perimeter, so the rollers need no rigid cross bars.
- Record the failure modes you did not fix. The team lists sheared axle bolts and a snapped deploy chain as open problems for a similar future intake.
- Compare the CAD with the built robot. The team states where they differ: hand-drilled bellypan holes, shorter shooter shafts, and climber tensioning by surgical tubing.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video

■ **Study pick** **1778 Chill Out** Zappy Feet — USA-WA — **Local EPA 49 (#11)** 43-22

'Zappy Feet': shooter with 2 bottom rollers and 1 top; amp bar on a virtual four-bar with no extra motor

SHOOTER (SPEAKER) Shooter with two rollers on the bottom and one on the top. Shooter compression is 1/2 in, with no horizontal compression. The team added the second bottom roller because a one-roller prototype shot above the amp bar; it also made intaking faster. Motors and ratios: Unconfirmed (see CAD).	**INTAKE** Intake compression is 0.2 in. The two bottom shooter rollers help bend the note into the shooter when intaking, per the OP. Intake type and motors: Unconfirmed (see CAD).
FEEDER / AMP / TRAP Amp scoring uses an amp bar on a virtual four-bar, which removed the need for an extra motor on the shooter subsystem. The design lead added it after the team tested an idea from RI3D. The reason was concern that notes would fall out of the amp, possibly into the bellypan. Trap: Unconfirmed (not stated by the team).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines: Unconfirmed (see code). Autonomous Award at PNW District Bonney Lake.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A linkage can replace an actuator: the amp bar on a virtual four-bar removed an extra motor from the shooter subsystem.
- A prototype result can set roller layout: a one-roller shooter shot above the amp bar, so the team added a second bottom roller.
- Write down compression: the team states 1/2 in shooter compression, no horizontal compression, and 0.2 in at the intake.
- Process before CAD: the team set priorities and strategy first, sketched several archetypes, and narrowed to one; the final robot stayed close to the first KrayonCAD.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub)

■ Study pick

1706 Ratchet Rockers Riot

USA-MO

Local EPA 48 (#16)

44-4

'Riot': swerve; belt-driven elevator carrying the amp/trap mechanism; rack-and-pinion shooter hood; Thrifty climber kit

SHOOTER (SPEAKER) The shooter hood moves on a rack and pinion. It ran on a P term only; the team states it should add an arm feedforward (kG/kS) because the hood has a swinging CG. The rack skipped teeth until the team increased the DP mid-season and changed to a steel shaft (drop-in, 3x stiffness). Flywheel motors and ratios are unconfirmed (see CAD).	**INTAKE** Unconfirmed (not stated by the team in the public release; see CAD).
FEEDER / AMP / TRAP An elevator carries the amp and trap mechanism. The two sides are belt driven with no mechanical link between them, so the team tuned belt turnbuckles to stop racking and skipped teeth. Control uses the WPILib ElevatorFeedForward and TrapezoidProfile classes. The team changed the elevator to NEO 550s after a failure at Roboteer Rumble (NEO encoder wires and a melted pulley).	**DRIVETRAIN** Swerve: a reply praised the swerve and the team's release poster answered 'Yeah we love it!'. Module type and ratios are unconfirmed (see CAD). A reply noted plate construction in the bracing and cross members; the team said these parts were machined on its Tormach PNC 770. The team reports zero structural failures anywhere on the robot all season.
ENDGAME The climber is the Thrifty climber kit; a team member confirmed this in the reveal thread. For the trap, the driver climbs and then holds Y to drive the elevator to max height, where the motor stalls at full extension. The trap code is not in the public repository.	**AUTONOMOUS** Autonomous Award at the St. Louis Regional. Routines and note counts are unconfirmed (see code).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Unlinked elevator sides need tuning: with no shaft tying the two belt-driven sides together, the team adjusted belt turnbuckles so the elevator would not rack or skip teeth, and states it would design this differently.
- A rack-and-pinion hood can skip teeth; the team fixed it mid-season by increasing the DP and changing to a steel shaft with 3x the stiffness.
- Bushings can replace bearings at moderate speed: the team believes most ratios are about 3:1 off the motors, 'roughly 1500-2000' (no unit given), with axial float, no tight press fits and oil worked in after assembly. The team replaced no bushings all season.
- The elevator runs WPILib ElevatorFeedForward and TrapezoidProfile instead of Smart Motion; the team states it would use MotionMagic on a Kraken.
- The team machines its bracing and cross member plates in-house and reports zero structural failures anywhere on the robot all season.

Onshape CAD: Onshape CAD · 2024 CAD release (CD) · Onshape CAD · Code (GitHub) · Reveal 'Riot' (CD) · Reveal video

Profile	2075 Enigma Robotics Masquerade USA-MI	Local EPA 48 (#15) 46-16-2

'Masquerade': note shooter with a shuttling shot mode; intake deploys from the start configuration; Trap added mid-season

SHOOTER (SPEAKER) Note shooter with a shuttling shot mode added before Kentwood. Shooter type, motors and ratios unconfirmed (no CAD released).	**INTAKE** The reveal photo caption shows the intake releasing from its start-of-match configuration. The team reports improved intake reliability before Kentwood. Rollers and motors unconfirmed.
FEEDER / AMP / TRAP Trap scoring was added before Kentwood; the trap and amp hardware is not described. Unconfirmed (not stated in the public release).	**DRIVETRAIN** Drivetrain unconfirmed (no CAD released).
ENDGAME The team redesigned the whole climb system before Kentwood and added Trap scoring. The team reports 9 of 12 Traps in MSC qualifications and 4 of 4 in playoffs. Harmony unconfirmed.	**AUTONOMOUS** Before MSC the team added new center autos and tuned automatic note pickup and auto localization. Note counts unconfirmed.
VISION Limelight 3G (LL3G), adopted before FIM Kentwood per a team update. Note detection drives automatic pickup. Localization during autonomous was tuned before MSC.	**PUBLIC CAD / DESIGN** Reveal thread with season updates (CD)

LESSONS FOR STUDENTS

- Mid-season changes can be large: between week 1 and week 5 the team switched to LL3G vision, redesigned the climb and added a Trap.
- Track mechanism success rates: the team reports 9 of 12 Traps in MSC qualifications and 4 of 4 in playoffs.
- Note detection for automatic pickup was added in season, then tuned before MSC.

CD thread: Reveal thread with season updates (CD) · Reveal thread with season updates (CD) · Reveal video · Video linked in the reveal thread

Profile	4481 Team Rembrandts CODA Netherlands	Local EPA 48 (#14) 39-10

'CODA': REV MAXSwerve; two-sided under-bumper intake; one shooter/amp 'Shamper'; dual climbers; 7-note auto

SHOOTER (SPEAKER) Combined amp and speaker scoring system that the team calls 'The Shamper'. The team states it combines a shooter with an amping mechanism. Rollers run on individual RPM setpoints. Auto aim uses lookup tables; the team states it chose them because they are easy to adjust live. For moving shots, a team member writes he is '99% certain' the code compensated for robot velocity by using a position further 'into the future'. Motors and ratios: Unconfirmed (not stated in the public posts).	**INTAKE** Two-sided under-the-bumper intake, per the reveal. Roller layout and motors: Unconfirmed (not stated in the public posts).
FEEDER / AMP / TRAP Amp scoring uses the same 'Shamper' system as the speaker shot, per the reveal. Indexer layout and trap mechanism: Unconfirmed (not stated in the public posts).	**DRIVETRAIN** REV MAXSwerve modules, per the reveal. The team updated 'all our SPARK Flex controllers' in April 2024; the post does not say which subsystems use them. Drive motors and gear ratio: Unconfirmed (not stated in the public posts).
ENDGAME Dual climbers, per the reveal. Trap and harmony: Unconfirmed (not stated in the public posts).	**AUTONOMOUS** 7-note auto, tested in April 2024 before the Championship. Changes to roller setpoints, Shamper wheels and grip tape did not improve long shots. The team then moved the centerline shooting point toward the center of the stage. In 8 timed test runs, 2 scored 7 notes and 6 scored 6. The team states it is not 100% consistent.
VISION PhotonVision. The default WPILib SwerveDrivePoseEstimator fuses vision tag readings with odometry, using vision standard deviations. The OP thinks it uses odometry until a tag is detected and the pose difference is not too large, and adds that he could be wrong. Camera hardware: Unconfirmed.	**PUBLIC CAD / DESIGN** 2024 Open Alliance build thread (CD)

LESSONS FOR STUDENTS

- One system, 'The Shamper', scores in both the amp and the speaker. Compare it with robots that use a separate amp mechanism.
- The 7-note auto post shows a test method: 8 timed runs with the same notes, each run listed with notes scored and a reason where a run fell short. Students can copy the table format.
- When shooter changes did not fix long auto shots, the team moved the shot closer and accepted the time cost. The post states the trade.
- The team logs sensor data, pose, camera readings, subsystem states and controller inputs, and plots current draw per motor per state in AdvantageScope to prevent brownouts.
- Auto aim uses lookup tables, and a team member believes moving shots used a projected future position. He states he prefers a tunable method over one that works better on paper.

CD thread: 2024 Open Alliance build thread (CD) · 2024 Open Alliance build thread (CD) · Reveal 'CODA' (CD) · Reveal video · State machines post (CD) · Code (GitHub org)

Profile	368 Team Kika Mana USA-HI	Local EPA 46 (#23) 40-5

Speaker shooter with split wheels, hook climbers and a trap mechanism; drivetrain unconfirmed

SHOOTER (SPEAKER) Speaker shooter with split wheels. The OP states the split wheels put a little spin on the note to stabilize the shot, an idea taken from an Open Alliance build. After Los Angeles the team added a second shooter motor and raised wheel speed for a more accurate wing-line shot, mostly for auto. Motor type unconfirmed (not stated in the public release).	**INTAKE** Intake type unconfirmed (not stated in the public release). After Los Angeles the team rebuilt the front end and made new bumpers. The OP states the intake was no match for the stage.
FEEDER / AMP / TRAP The robot scores in the trap (see endgame). Feeder and amp detail unconfirmed (not stated in the public release).	**DRIVETRAIN** Drivetrain unconfirmed (not stated in the public release).
ENDGAME Hook climbers on the stage chain and a trap mechanism. After Los Angeles the team pre-deployed the hook climbers and raised speed by 20%, aiming for a trap under 3 seconds.	**AUTONOMOUS** The team added auto modes after Los Angeles to fit with alliance partners. Autonomous Award at the Hawaii Regional. Note counts unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Reveal thread with spec sheet (CD)

LESSONS FOR STUDENTS

- Iterate between events with a short change list: the team named four changes after Los Angeles (climber pre-deploy with a 20% speed increase, a second shooter motor, more auto modes, a more robust front end with new bumpers).
- Split shooter wheels add spin to stabilize the note; the OP credits the idea to an Open Alliance build.
- Design the front end for contact with field structures: the OP states the intake 'was no match for the stage', and the team made the front end more robust.
- No CAD or code is public; the spec sheet PDF and the reveal video are the study material.

CD thread: Reveal thread with spec sheet (CD) · Reveal thread with spec sheet (CD) · Reveal video

■ Study pick

2910 Jack in the Bot Typhoon

USA-WA

Local EPA 46 (#22)

50-16-1

'Typhoon': +/- 360 deg turret shooter; two-entry feed loads notes at any turret angle; about 16.5 in tall; no Trap

SHOOTER (SPEAKER) Turret shooter. A team member stated the turret rotates +/- 360 degrees and loads notes from any angle in that range. The robot is about 16.5 in tall with the shooter down. Shooter motors and ratios unconfirmed (see CAD and binder).	**INTAKE** The team states the intake is modeled after the Spectrum intake and picks up notes from many angles. Motors and geometry unconfirmed (see CAD).
FEEDER / AMP / TRAP Load-from-any-angle feed. The team compares it to the 254 spindexer with two entry points: the note enters from the 180 degree arc facing the intake. Feeding takes slightly longer with the turret at 90 or 270 degrees, because the teacup rollers contact the note as it enters the indexing zone. The team did not design in a Trap function. At reveal, a team member wrote that the amp plan still had issues to resolve.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Changed after the reveal. The first plan was a spring-powered polycarbonate hook released by a solenoid; the pivot would then pull down to climb. In testing, the pivot motors in brake mode could not hold the robot up long enough after the match. The hook sat further from the center of gravity than planned, so the robot hung at a steep angle. The turret motor brake could not hold the turret, so the base could rotate and might touch the ground (team post 2024-04-13). The released robot uses a climber gearbox that put a high load on its motor. After some successful climbs the motor stalled at low voltage and current. The team believed new motor protection firmware caused this. After a firmware rollback the motor drew about 180 amps, and one burned out at DCMP even with an e-stop after each climb. Without a root cause, the team recommends other teams explore alternate designs.	**AUTONOMOUS** Routines unconfirmed (see code).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Study how a turret accepts notes at any angle: a spindexer-style feed with two entry points takes the note from the 180 degree arc facing the intake.
- Measure the cost of the worst-case angle: the team states feeding is slightly slower at 90 and 270 degrees, where the teacup rollers touch the note first.
- Check holding torque before you rely on motor brake mode. In tests of the first climb plan, brake mode on the pivot motors could not hold the robot up, and brake mode on the turret motor could not hold the turret.
- Size climber motors with margin. The team's climber stalled at low current, which the team believed was new motor protection firmware. After a firmware rollback it drew about 180 amps, and a motor burned out at DCMP.
- Publish unresolved failures with the CAD: the team released the climber problem without a root cause and advised other teams to try other designs.

Onshape CAD: Onshape CAD · CAD, code and tech binder release (CD) · Onshape CAD · Code (GitHub) · Tech binder (Google Drive) · Robot reveal (CD) · Reveal video

Profile	33 Killer Bees Buzz XXIX USA-MI	Local EPA 45 (#30) 49-17-2

'Buzz XXIX': 4x Kraken X60 shooter with independent wheel sets, full-width compliant intake, side chain climb

SHOOTER (SPEAKER) A new shooter ran at Livonia, the Michigan State Championship and the Championship. It uses 4x Kraken X60, with each set of wheels driven independently. A polycarbonate, Teflon-skinned tab-and-slot box compresses the note side to side to 12 in. The team says this increased consistency when shooting from back at the wing line.	**INTAKE** Full-width compliant front roller. It replaced vertical flappers that worked in the shop; after week 1, the team reports that new notes on new carpet slid 'like hockey pucks on ice'. Motor count unconfirmed (not stated by the team).
FEEDER / AMP / TRAP Feeder and amp mechanism unconfirmed (not stated in the public release). The team had planned to score in the trap and dropped it right before the Kettering event.	**DRIVETRAIN** Drivetrain unconfirmed (not stated in the public release).
ENDGAME Side chain climb, added for Livonia. A mast similar to the team's 2020 robot carries a spring-loaded hook that pops out and grabs a single chain link. The team ties the climb to ranking higher after it dropped the trap.	**AUTONOMOUS** Routines and note count unconfirmed (not stated in the public release). The team reports that in some Championship matches the auto path ended behind the truss and the robot still made its shots.
VISION Limelight cameras. A fourth Limelight, mounted in the center of the robot, was added for the Michigan State Championship for TargetPose. The team reports that targeting and localization software changes between MSC and the Championship increased shot accuracy in auto and teleop.	**PUBLIC CAD / DESIGN** Robot reveal and update thread (CD)

LESSONS FOR STUDENTS

- A shooter box that compresses the note side to side to 12 in, with independently driven wheel sets, is the team's stated change for consistent shots from the wing line.
- Test intakes on worn and new field conditions: flappers that worked in the shop failed with new notes on new carpet, and a full-width compliant roller replaced them.
- Scope can change mid-season: the team dropped the trap before its first event and added a fast side climb to rank higher.
- The team reports that a fourth Limelight and late software work on localization raised shot accuracy between MSC and the Championship.

CD thread: Robot reveal and update thread (CD) · Robot reveal and update thread (CD)

■ Study pick

3467 Windham Windup Nocturne

USA-NH

Local EPA 45 (#28)

59-22-1

'Nocturne': arm and shooter, harmony climb hooks, leaf-blower trap assist under the shooter; drivetrain unconfirmed (see CAD)

SHOOTER (SPEAKER) The trap post names an arm with uprights and a shooter above the blower. Whether the shooter rides on the arm is unconfirmed (see CAD). The team states its cleanup play often relied on the subwoofer shot. The team states localized passing was a late addition. Flywheel type, motors and ratios unconfirmed (see CAD).	**INTAKE** Intake type unconfirmed (see CAD). The robot browned out at the end of two Championship playoff matches. The team states that front-court play ran the shooter and intake motors much more than cycling did.
FEEDER / AMP / TRAP Trap: a converted Milwaukee M12 Compact Spot Blower blows the trap door open so the robot can shoot a note into the trap. The team took the concept from 4028's reveal. It replaced the blower's non-FIRST-legal 775 motor with a 775 Pro and press-fit the original output shaft onto it. At the Championship the blower trap scored in a practice match but not in a qualification match. The team states it worked no better at the NE District Championship. Feeder and amp details unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). In December 2023 the team planned to run SDS MK4i swerve modules again, with swerve code derived from Team 364's BaseFalconSwerve. The Championship recap names a 'Front Left Steer' CAN device in the RIO logs.
ENDGAME The robot used harmony climb hooks. At the Championship it harmony climbed three times: QM71 and both the original and replay of QM110, which earned a ranking point. On kickoff day the team listed the trap as TBD on its priority list. The blower was on the robot from Greater Boston onward. Climb mechanism details unconfirmed (see CAD).	**AUTONOMOUS** Rush autos to the centerline notes. In Championship playoff matches the team ran a 'Poop n' Scoot' auto from the Amp side that reached the centerline note in about 2.33 seconds. The team chose the next note from its prediction of the opponents' autos. The team states these autos were fast but not always accurate. Autonomous Award at North Shore and at the NE District Championship Ganson Division.
VISION The team tested the blower trap shot with auto align. Its next step was to improve AprilTag pose estimation accuracy for shot repeatability. Camera hardware unconfirmed. In December 2023 the team planned AprilTag work with one or more Limelights.	**PUBLIC CAD / DESIGN** Full robot CAD (Onshape)

LESSONS FOR STUDENTS

- A trap mechanism from a power tool: a Milwaukee M12 blower with its 775 motor swapped for a FIRST-legal 775 Pro.
- Adopting another team's idea in season: after 4028's reveal, the team tested blower mounting angles at the practice field and then mounted it between the arm uprights.
- Model purchased parts at true size: the team states the CAD blower matches the length and diameter of the blower with its twist-on extension.
- Plan for wear: the team states Nocturne struggled to stay on the field from the end of the NE Championship. By the off-season it had played a little over 110 matches. It then built 'Black Cherry', a clone of 1678's 2024 robot, to practice an elevator and an over-bumper intake.
- Leave slack on every CAN wire: in a Championship playoff match a wire with a snapped Molex SL crimp clip, routed under the arm gearbox with little slack, made the robot stop moving at intervals. The team found it from a CAN Stale Error in the RIO logs.

Onshape CAD: Full robot CAD (Onshape) · Full robot CAD (Onshape) · Trap blower thread (CD) · 2024 build blog with Championship recap (CD) · Off-season 'Black Cherry' write-up (CD)

Other profiled robots

Each has public CAD, a binder, code or a detailed release thread. Sorted by Local EPA. ■ marks the study picks.

■ Study pick **1114 Simbotics** Simbot Skyfall
Canada

Local EPA 45 (#32)
35-28

'Simbot Skyfall' 2.0: rebuilt shooter that amps out the rear via a roller arm; chain climb with a latch; drivetrain unconfirmed

SHOOTER (SPEAKER) Version 1.0 used a sliding shooter. The team states the slide slowed cycle times and the shooter lacked accuracy. Before McMaster the team redesigned and rebuilt the shooter and the climber. Barlow helped make new sheet metal parts for the redesigned robot. Motors and ratios are unconfirmed (see CAD).	**INTAKE** The team states it used the same outer-roller-faster gearing in its original intake system. Other intake detail is unconfirmed (see CAD).
FEEDER / AMP / TRAP 2.0 scores the amp out of the rear of the shooter through an arm. The note turns inside the arm, so the outer rollers need a higher surface speed than the inner ones; the team gives 3.11:1 between top and bottom roller. A light sensor in the arm lets the robot partly eject the note, align it with the amp opening, then eject it fully. Amp geometry came from sketches, not a prototype. Trap capability unconfirmed.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Stage chain climb. The team states the 1.0 climber was too complex to give consistent results, and 2.0 replaced it. The 2.0 arm did not stay up long enough for the climb to count, so team members held an internal contest to design a latch, and the winning design went on the robot. The arm must swing onto the chain at a tuned speed for the latch to engage.	**AUTONOMOUS** The team cites a Milstein playoff match with 4 notes scored in auto (30 in total). Paths and path tool are unconfirmed.
VISION After the Ontario Provincial Championship the team added vision tracking that detects game pieces and corrects the robot's path to them in autonomous and teleop. Camera and software are unconfirmed.	**PUBLIC CAD / DESIGN** Simbot Skyfall 2.0 CAD (Onshape)

LESSONS FOR STUDENTS

- Two released versions of one robot: compare 1.0 and 2.0 against the team's stated reasons for the rebuild (slow sliding shooter, low accuracy, complex climber).
- When a note turns through a roller path, run the outer rollers faster; the team used 3.11:1 between top and bottom roller to stop jams.
- A light sensor can stage a note for the amp: partly eject, align with the opening, then eject fully.
- A climb that does not hold can be fixed with a latch; this latch needs the arm to swing onto the chain at a tuned speed.
- With no time to prototype, the team set amp geometry from sketches and from angles and wheel spacing referenced from other teams; it credits teams 33, 5460 and 3467 for the redesign.

Onshape CAD: Simbot Skyfall 2.0 CAD (Onshape) · 2024 CAD release (CD) · Simbot Skyfall 1.0 CAD (Onshape) · Simbot Skyfall 2.0 CAD (Onshape) · 2024 code release (CD) · Code (GitHub)

■ Study pick

3061 Huskie Robotics Artemis

USA-IL

Local EPA 42 (#66)

32-16-2

'Artemis': swerve, intake with IR note sensing, kicker into a pivoting flywheel shooter; 92 lb at Heartland inspection

SHOOTER (SPEAKER)
Kickoff requirements: score in the amp, speaker and trap; adjustable angle greater than 90 degrees; shoot amp and speaker from against the subwoofer. The competition shooter has flywheels on a pivot driven through a gearbox, with a CANcoder after the gearbox to read shooter position. The team added a mechanical hardstop to back up the software pivot limit and changed its plates in the pit at Heartland. At Heartland it also chamfered battery handles that hit the flywheels with the shooter all the way down. Shooter motors and ratios are unconfirmed (see CAD).

INTAKE
Kickoff requirement: an under-the-bumper intake, double-sided on opposite sides, 'touch it own it'. Week 1 prototypes tested 2 in stealth wheels, a grip-taped hex shaft with a polycarbonate ramp, and sushi rollers; the hex shaft and ramp worked better than the wheels, but the note stuck on the polycarbonate. The competition intake is a removable module with IR sensors for note detection; the team swapped in a spare at Heartland to replace worn wheels. The shooter hardstop and pivot made the swap harder than planned, so the team plans snap-on spacers so the intake can collapse for removal.

FEEDER / AMP / TRAP
Week 1 ideation lifted notes into a central area and flipped them around a drum to stage them for the shooter; whether the final robot uses the drum is unconfirmed (see CAD). The Heartland recap names a kicker between the intake and the shooter, and an inner rail that guides notes up into the shooter. Notes sometimes stuck on that rail, so the team added a 3D-printed part and later replaced it with a smoother print. The kickoff plan scored amp and trap with the shooter; the final amp and trap method is unconfirmed (see CAD).

DRIVETRAIN
Kickoff requirement: MK4i swerve drive, as fast as possible, with clearance for notes. The Heartland recap confirms swerve modules with tread, Falcon motors on the modules (the recap mentions soldered Falcon wires during a module swap), and a spare module the team swapped in after a steering motor failed in match 60. The front rail bent during defense and was replaced; the electrical board mounts upside down under a removable bottom polycarbonate panel. Final module model is unconfirmed (see Drivetrain CAD).

ENDGAME
Kickoff requirement: support the robot on the chain with at least 1 other robot and raise it off the ground. Trap scoring was a shooter requirement. A separate Climber CAD document is linked, and the week 1 climber update links the WCP GreyT Telescope product page. Final climber hardware is unconfirmed (see CAD).

AUTONOMOUS
3061-lib v2024.0.0 adds a shouldFlipAutoPath method that PathPlanner calls before an auto to flip the path across the field center, and a tuning auto for swerve rotation. In its 2025 build thread the team says it ended 2024 autos with a sprint and the swerve modules in coast mode, to travel from the centerline toward the source between auto and teleop. Note counts for the 2024 autos are unconfirmed.

VISION
A Limelight is on the competition robot. Its first mount let the bumper block a third of the view during intake, so the team printed a new mount at Heartland. 3061-lib v2024.0.0, the base for the 2024 code, supports multiple PhotonVision cameras for pose estimation using PhotonVision's MULTI_TAG_PNP_ON_COPROCESSOR feature. Camera count on the robot is unconfirmed.

PUBLIC CAD / DESIGN
2024 full robot CAD (Onshape)

LESSONS FOR STUDENTS

- Each subsystem (climber, drivetrain, intake, shooter) is a separate Onshape document, so one mechanism can be studied without loading the full robot.
- The team wrote per-subsystem design requirements at kickoff, for example a 2' 3.5" max height and a shooter angle range greater than 90 degrees, and designed against them.
- When they found no public note friction data, they measured it: each material was tilted under a note until it slipped, and the angle was recorded.
- The Heartland recap pairs each failure with a short-term pit fix and a longer-term fix, for example a spare swerve module with its own CAN IDs for faster swaps.
- A pivot with backlash can still be measured: the team placed the CANcoder after the shooter pivot gearbox and added a mechanical hardstop behind the software limit.
- 3061-lib holds game-agnostic features (AdvantageKit logging and replay, PhotonVision pose estimation, drive-to-pose), and frc-software-2024 holds the Crescendo code.

Onshape CAD: 2024 full robot CAD (Onshape) · 2024 build thread (CD) · 2024 full robot CAD (Onshape) · Climber CAD (Onshape) · Drivetrain CAD (Onshape) · Intake CAD (Onshape) · Shooter CAD (Onshape) · 2024 code (GitHub) · 3061-lib v2024.0.0 release (CD)

■ Study pick

581 Blazing Bulldogs Titan

USA-CA

Local EPA 40 (#78)

46-16

'Titan': MK4i swerve; pivoting split-flywheel shooter; angled amp/trap elevator; chain-driven climb hooks with trap

SHOOTER (SPEAKER)
Two Kraken X60 motors power eight 4 in OD polycarbonate flywheels wrapped in CatTongue grip tape. Miniature gearboxes reverse the lower flywheels, so one Kraken X60 drives each side. The left and right sides run independently to put horizontal spin on the NOTE; the flywheels spin on dead axles. One Kraken X60 drives the pivot through 3D printed Onyx herringbone sector gears.

INTAKE
Roller geometry was first modeled on the 6328 V20 intake. One Kraken X60 drives all rollers; geared shafts reverse direction, and 9 mm wide 5 mm HTD belts link the shafts. The first contact roller is 1/2 in fiberglass tube in 7/8 in OD surgical tubing, so it springs back after a hit. Bent 1/16 in HDPE plates and printed parts funnel the NOTE; beam-break sensors confirm the intake.

FEEDER / AMP / TRAP
Two 1 5/8 in flex wheel queuer rollers hold the NOTE. The team pulses them backwards a few times so a squashed NOTE can return to round. Amp and trap scoring use a one-stage angled elevator on one Kraken X60 at 6.25:1, with rollers on a second Kraken X60. Three top rollers dunk the NOTE; for the TRAP the rollers alternate voltage to 'jackhammer' it in.

DRIVETRAIN
Four SDS MK4i swerve modules, L3 gearing with 16t pinions, eight Kraken X60 motors. Frame perimeter 25.5 in x 28.5 in without the intake, 30 in x 28.5 in with it. Cut-outs in the frame let the elevator pass under the STAGE.

ENDGAME
Each hook is two 1/8 in aluminum plates with printed webbing. Bearing blocks ride #25H chain driven by two Kraken X60 at 15.6:1; one output shaft through both gearboxes keeps the hooks parallel. The 5-stage climb runs on two buttons: hooks engage, the amp elevator extends onto the STAGE on wall-ride wheels, hooks retract, the rollers score the TRAP, the amp mechanism partly retracts. Harmony unconfirmed.

AUTONOMOUS
Paths built in PathPlanner, with autos for each position around the SUBWOOFER. The team states an auto from any position can score six NOTES. If the beam breaks do not see a NOTE, the path skips to the next NOTE instead of shooting.

VISION
One Limelight 3G localizes the robot. Pose estimation sets coarse speaker aim; the TX angle to the speaker tag then aims the drivebase and the TY angle sets shooter angle and RPM. Feeding shots use pose estimation from all field tags. Near the amp-zone corner the robot angles the shot to bounce the NOTE into the speaker.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- A late redesign can work with a fixed schedule: the team states 4 days of design, 7 days of assembly and wiring, and 9 days of debug and practice before its first event.
- Read the trap climb as a mechanism and a sequence together: the amp elevator doubles as a brace against the STAGE, and two buttons drive all 5 stages.
- Friction is a design variable: the team reduced shooter pulley diameters at fixed belt lengths after it saw high current draw. It set intake belt center-to-center distances -.03 short after 6328 suggested it look at intake current draw.
- Miniature reversing gearboxes let one Kraken X60 drive both upper and lower flywheels on each side, and independent sides give NOTE spin.
- Two state machines run teleop: a note manager tracks the NOTE inside the robot, and a robot manager coordinates scoring, climbing and trapping.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video · Robot reveal (CD)

■ Study pick

4795 Eastbots Perseus

USA-NC

Local EPA 39 (#93)

47-13

'Perseus': 22 lb pivoting shooter/amp tower on 2x NEO Vortex at 144:1 with gas shocks; intake handoff; MAXSwerve

SHOOTER (SPEAKER) Shooter on a pivoting tower that also scores in the amp. The tower weighs 22 lbs. Two NEO Vortex motors drive the pivot through a 144:1 combined reduction, with two gas shocks for stability. In the January design review the team chose a horizontal-axle shooter, inspired by Team 95, with each side driven independently. The team chose it for better predicted consistency with warped notes and saw no significant benefit from added spin. Final flywheel motors and ratios unconfirmed (see CAD).	**INTAKE** The intake sits outside the swerve perimeter and inside the bumper perimeter. The team states a handoff allows intake in multiple positions. In January the team prototyped an over-the-bumper intake, sketched an 'in-between' intake and did not prototype an under-the-bumper intake. It kept the over-the-bumper and in-between concepts. In a December 2024 post the team wrote that its offseason intake worked 'maybe even better than Perseus's intake'.
FEEDER / AMP / TRAP The intake hands off to the pivoting tower. The same tower scores in the speaker and the amp. In the January design review the team preferred active control of a note until it is scored over passive amp ramps. Trap capability unconfirmed (see CAD).	**DRIVETRAIN** Swerve. In a December 2024 post the team wrote that it ran MAXSwerve for the past two seasons before moving to WCP Swerve X on its offseason robot. Frame size is 31 in x 26 in.
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** In January the team was building a Kitbot to place on a swerve drivebase so programmers could start path work early. Final routines unconfirmed (see code).
VISION Three AprilTag-detecting cameras; the code gathers and processes their data for robot positioning. The code also has a note tracking pipeline and a 'note relative' drive mode.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Use passive assist on a heavy pivot: two gas shocks support a 22 lbs tower driven by two NEO Vortex motors at 144:1.
- Combine encoders for pivot control: the code interpolates between an absolute and a relative encoder.
- One pivoting tower can serve two goals: the team uses the same tower for speaker shots and amp scoring.
- Package the intake between the swerve perimeter and the bumper perimeter, with a handoff that works in multiple positions.
- Pick a concept against a stated failure mode: the January review chose a horizontal-axle shooter because the team was concerned that note deformation and wear would vary compression in a vertical-axle shooter.

Onshape CAD: Onshape CAD · CAD, code and binder release post (CD) · Onshape CAD · Code (GitHub) · Tech binder (Google Drive) · 2024 Open Alliance build thread (CD)

■ Study pick

6329 The Bucks' Wrath Dial

USA-ME

Local EPA 39 (#95)

53-19-1

'Dial': high-pivot shooter that also scores amp and climbs; full-width under-bumper intake; MK4i swerve; polycarb superstructure

SHOOTER (SPEAKER) High-pivoted shooting system. The arm length is 12.447 in, set by other features of the arm system in the reference sketch. The lowest 'home' angle is 25 deg; the OP reports this angle limited shot distance and recommends going lower. Arm position uses only the Krakens' internal encoders: the arm starts each match folded down against TPU hard stops and is not re-homed during the match. The top of the shooter is open in the amp position; the team states no retention roller was needed.	**INTAKE** Full-width under-the-bumper intake. The first intake was over the bumper (test video from 2024-01-24); the team says it worked but did not center notes well and was not wide. Most teams it talked with that were going to Granite State used under-bumper intakes, so the team tested one on the same geometry and had it running on 2024-01-27. The 1x1 front tube bent in play. Before the Championship the team filled it with maple, added a 1x1.5 aluminum angle under it, and remade the bumpers in maple.
FEEDER / AMP / TRAP The shooter flips up to score in the amp; the public release does not describe a separate amp mechanism. Indexer and trap details unconfirmed (see CAD).	**DRIVETRAIN** SDS MK4i swerve modules with the L3 ratio and standard black nitrile SDS wheels (team post 2024-05-06).
ENDGAME The pivoting shooting system is also used to climb, at various spots on the chain. Trap and harmony unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Basic tx/ty tracking on the AprilTag under the speaker opening, used for shooting while moving. The team states it had no localization. Camera model unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD and tech binder

LESSONS FOR STUDENTS

- One pivot, three jobs: the high-pivot shooter scores the amp, shoots over defenders and climbs on the chain.
- Set the low end of a pivot range from the shots you need: the 25 deg home angle ended up limiting shot distance.
- Structure from material on hand: two sandwiched sets of 3/16 in polycarbonate plates with 3/8 in hex standoffs, tested on the prototype with aluminum as the fallback. The OP reports no plate broke or bent all season.
- Fix a bending member at its root: after the 1x1 intake front tube bent, the team filled it with maple, added an aluminum angle under it, and switched the bumpers to maple.
- 3D printing at scale: PolyMax PLA on a Bambu X1C, 4-6 walls, 30-50% infill and WCP 1/2 in hex inserts, with PC for parts that kept breaking in PLA. Printed herringbone gears run in pairs on each side for load sharing and redundancy.

Onshape CAD: Onshape CAD and tech binder · Reveal and CAD release thread (CD) · Onshape CAD and tech binder · Reveal video · Early over-bumper intake test (video)

■ Study pick

1787 The Flying Circuits Ronnie

USA-OH

Local EPA 38 (#123)

30-11

'Ronnie': pivoting flywheel 'accelerator' (1 Kraken per side); 'snake path' intake; passive 'slam dunk' hood for Amp and Trap

SHOOTER (SPEAKER)

The team calls the shooter an 'accelerator'. It pivots on an arm and has 4 split flywheel axles. In February the team joined the axles in the middle with a 3D-printed part and bearings instead of cantilevering them. Each side is driven 1:1 by one Kraken (team reply, 2024-03-22). The wheels were BaneBots 3-7/8 T81; after week 1 the team planned tests with 4 in AndyMark Stealth Wheels for more compression. For note spin the team changed from one side spinning faster to more wheels on one side. Before week 1 the Speaker shot was tuned from the subwoofer to a little behind the podium. After week 1 a mentor made a Desmos equation for the arm angle that accounts for the note's parabolic path. Side panels are 1/2 in polycarbonate. The off-season Noodle uses painted wood panels and one NEO per flywheel (2 per side).

INTAKE

Ronnie uses a 'snake path' intake that routes notes around to the accelerator and supports Trap scoring. Before week 1 the team reported fast pickup, but the transfer to the accelerator sometimes took longer than desired. At Greater Pittsburgh, notes taken off-center went through folded against one side. The planned fix was to remove the silicone from the center of the rollers. Noodle uses a direct path, and the team reports that notes seem to intake faster.

FEEDER / AMP / TRAP

Amp and Trap use a passive 'slam dunk' hood above the accelerator. The accelerator swings into the Amp or Trap, and the contact pushes the hood over the note exit, which turns the note down. The team added a string-and-pulley system to pull the hood down past a set angle. At Greater Pittsburgh the string came loose in one match and two Amp shots flew upward; the team was working on 3D-printed pulleys for the string as a fix. A note that could not leave the hood also stalled the flywheels and jammed the accelerator for the rest of a match, so the team planned to reduce hood compression. Noodle is a Speaker and Amp only robot.

DRIVETRAIN

Swerve. The team's 2024-01-09 intake discussion refers to the placement of the swerve modules. Module type unconfirmed (see CAD).

ENDGAME

Chain climb on the Stage, with a planned Trap shot. The team wrote that the climber would lift the robot high enough for the hood to face the Trap, so the Trap shot would work like the Amp shot. Before week 1 the robot climbed the chain, but the Trap shot still needed tuning. At Greater Pittsburgh the climb was manual and slow: the driver stepped the arm through setpoints, then moved the hooks. The team found it cannot lower the arm after climbing, or the robot tips forward. Planned changes were a 'quick climb' with the accelerator down, a latch to hold the robot up after the match, a reshaped hood to open the Trap door, and an extra roller at the bottom of the hood.

AUTONOMOUS

Before week 1 the robot ran three different three-note autos, one starting near each face of the subwoofer. After Greater Pittsburgh the team set a goal of at least a 5-piece auto, and a 6-piece auto if possible, for the Buckeye Regional. Final routines unconfirmed (see code).

VISION

Unconfirmed (not stated in the public release). In the 2023 off-season the programmers started custom AprilTag localization software. After week 1 the team planned note tracking, at least for autonomous, to correct paths when a pickup misses.

PUBLIC CAD / DESIGN

Onshape CAD

LESSONS FOR STUDENTS

- A passive mechanism can replace an Amp/Trap actuator: Ronnie's hood is pushed into place by contact when the accelerator swings into the Amp or Trap.
- Write down failure modes after the first event: the team found a loose hood string and a hood that stalled the flywheels by over-compressing the note, and planned a fix for each.
- Model the note's arc: misses at week 1 led the team to replace a straight-line aim model with a Desmos equation for a parabolic path.
- One requirement shapes the note path: the Trap led to Ronnie's longer 'snake path' intake, and dropping it on Noodle allowed a direct path that the team reports intakes faster.
- Motor and material choices can change between versions of one design: Ronnie uses one Kraken per shooter side and 1/2 in polycarbonate side panels; Noodle uses one NEO per flywheel and painted wood.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 Open Alliance thread (CD) · 2024 code (GitHub) · 2024 Open Alliance thread post (CD) · Off-season reveal 'Noodle 3.0' (CD) · Noodle photos and videos (Google Drive)

■ Study pick

5951 Makers Assemble

Off-season robot (name not stated); competition robot: MAroon

Israel

Local EPA 38 (#113)

43-21-2

Off-season 2056-concept robot (CAD): under-bumper intake, pivot arm, fixed-speed shooter, no climber; swerve

SHOOTER (SPEAKER) Fixed-speed shooter: the team shoots at a static shooter speed and adjusts only the angle, from a linear interpolation table. The team widened the space between the polycarbonate shooter walls after notes stuck. At high speed the shooter wheels widened and rubbed a support rod; the team was moving the support. The shooter beam break sensor was unreliable in some lighting.	**INTAKE** Under-the-bumper roller intake, following the 2056 concept. One prototype tested passive centering on one side and active centering on the other; the team kept passive. Roller pulley slack was fixed with full-length shafts.
FEEDER / AMP / TRAP Pivot arm with an 'arm feeder' on sushi rollers; the team respaced the rollers to remove a dead spot. The team prototyped the arm angle for amp scoring. Arm backlash was reduced with oversized shafts in the arm gearbox and Loctite 648. A limit switch was added for consistent arm zeroing.	**DRIVETRAIN** Swerve, 29 in x 31 in frame. On two modules the team lifted the motors to widen the intake. It lifted both motors on each of those modules, not one, so the motors stay uniform and easier to work with. Module type unconfirmed (see CAD).
ENDGAME No climber. The team states it opted out of climbing because of the added complexity and limited strategic impact.	**AUTONOMOUS** Two auto routines running as of 2024-10-16, with a third planned. Note counts not stated in the text.
VISION Limelight; the team repositioned its stand for a better view. Software includes auto angle alignment for the amp and a 'stop and shoot' feature. The team reported odometry jumps that occasionally caused missed shots.	**PUBLIC CAD / DESIGN** Off-season robot CAD (STEP, Google Drive)

LESSONS FOR STUDENTS

- Build from a published binder: the robot is a 2056 concept scaled to the team's capabilities, using 2056's technical binder and drawing package.
- Test a design choice with one prototype: passive centering on one side, active on the other; passive worked and was kept.
- Trade module symmetry for intake width: on two swerve modules the team lifted the motors to widen the under-bumper intake, lifting both motors on each so they stay uniform.
- Arm backlash: oversized shafts and Loctite 648 in the arm gearbox, plus a limit switch for consistent arm zeroing.
- Software structure: after the season the team moved to a state machine and an IO layer similar to 6328's, which made it easier to test code without the robot.

Drive folder / doc: Off-season robot CAD (STEP, Google Drive) · Open Alliance thread with off-season build and CAD release (CD) · Off-season robot CAD (STEP, Google Drive) · 2024 Open Alliance build thread (CD) · 'MAroon' reveal thread (CD) · Code (GitHub organization)

■ Study pick

5419 Berkelium Radium the Roomba

USA-CA

Local EPA 37 (#132)

26-14-1

'Radium the Roomba': swerve; 4-side ground pickup via street sweepers; elevator + arm shooter; Amp, Trap and chain climb

SHOOTER (SPEAKER) Variable-angle shooter. The arm sets the shot angle and is fixed to the shooter at one angle. The code auto-aims into the Speaker from anywhere in front of the wing line, with full-speed, velocity-compensated auto shoot. Motors and ratios are unconfirmed (see CAD and binder).	**INTAKE** Under-bumper intake that picks up from the front and back and sits under and in the belly pan, protected from collisions. Compliant star-wheel 'street sweepers' sweep notes toward it, which gives ground pickup from all 4 sides. The team moved the sweepers inward and cut them down to stay inside the frame perimeter, and drove the two sides independently because the poly cord did not transfer torque well. A planned polybelt feeder was dropped as a degree of freedom with minimal return.
FEEDER / AMP / TRAP An elevator and arm carry the shooter. The elevator reaches the Trap, gives a downward angle on the Amp shot for consistency, and lets the robot shoot over defense. The arm is light and fits inside the moving stage of the elevator. The code runs an automated note pickup and handoff sequence.	**DRIVETRAIN** Swerve drive (stated in the release post). Module type and ratio are unconfirmed (see CAD).
ENDGAME Climbs on the chain and scores in the Trap. One-way spring-loaded hooks on mini-carriages let a single elevator stage climb twice as high. A team member confirmed the hooks use the return-side rigging of the elevator. The robot drives under the Stage; harmony is unconfirmed.	**AUTONOMOUS** Note count unconfirmed. The team states 4-side pickup is a key speed feature in auto. Driving the sweeper sides independently lets the robot pass notes under itself for more auto options.
VISION One camera running PhotonVision. Its pose is fused with swerve odometry for field localization.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Intake geometry can widen pickup without a second intake. Compliant star-wheel street sweepers feed a front-and-back intake, so the robot picks up from all 4 sides.
- Count the degrees of freedom you add. The team dropped a planned polybelt feeder under the robot because the sweepers already moved notes to the intake.
- The return side of elevator rigging can do work. One-way spring-loaded hooks on it let a single stage climb twice as high.
- Test intakes with worn game pieces. The team reports that notes sometimes stuck under the robot, especially older used notes.
- The code lets the driver queue one command in advance to cut transition time between states.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Technical binder (Google Drive) · Code (GitHub) · Robot reveal (CD)

■ **Study pick** | **131 CHAOS** Double Time | USA-NH | **Local EPA 36 (#144)** | 45-25-1

'Double Time': under-bumper intake; lift with chain-pull notches for the climb; launcher scores Speaker, Amp and Trap

SHOOTER (SPEAKER) The launcher centers and indexes notes, launches them, and scores in the Amp and Trap. The team designed it to put spin on the note but never used that feature. Motors and ratios are unconfirmed (see CAD).	**INTAKE** Under-the-bumper intake, packaged in the drive subsystem with the bumpers and electronics. The team first dismissed an under-the-bumper intake. They changed after 95's build thread showed them how much harder an over-the-bumper intake made integration.
FEEDER / AMP / TRAP Indexing, Amp and Trap scoring all happen in the launcher. Changed between Week 0 and Week 1: the outer roller went onto its own motor so the robot could score in the Trap by pushing the note out the top. Before that, one NEO Vortex drove the outer roller and the other rollers. After the team fixed height issues seen at Week 0, it no longer needed the top path. It kept the separate roller because it lets the robot take a note directly from a human player.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The 2024 release post does not state the drivetrain type.
ENDGAME The lift includes notches that pull the Stage chain down to start a climb. The launcher scores in the Trap. In the reveal thread the team states that Week 0 taught them about 'vertical efficiency'. Harmony is unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release). The team code is public on GitHub.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Adapt a lift design to the parts you can buy. 131 could not get the WCP inline bearing blocks, so they designed for clamping blocks and moved the pulley supports inside the tubing to keep full carriage travel.
- One mechanism can do two jobs: notches on the lift pull the Stage chain down to start the climb.
- Read other teams' build threads during the season. 131 changed to an under-the-bumper intake after 95's build thread showed the integration cost of an over-the-bumper intake.
- Putting one roller on its own motor added a Trap scoring path out the top of the launcher and a human-player intake path.
- The team states that attending Week 0 got them mechanically complete early, which left time for software features that reduced driver workload.

Onshape CAD: Onshape CAD · 2024 CAD release (CD) · Onshape CAD · Code (GitHub) · Robot reveal (CD)

■ Study pick

3357 COMETS Vaisala

USA-MI

Local EPA 36 (#150)

47-15

'Vaisala': 24x28 in, 75 lb swerve robot; roller-pinion sector gears; silicone-over-tube rollers; 14 SPARK MAX

SHOOTER (SPEAKER) Shooter fed by a shared intake/index/shooter system that uses 4 motors. The OP calls the shooter deck probably the team's most dense set of mechanisms. The robot also has a pivot on 1 motor and sector gears with a roller-pinion profile; which joint each drives is unconfirmed (see CAD). Each motor has a SPARK MAX controller; motor types and wheel details unconfirmed.	**INTAKE** Rollers are polycarbonate or aluminum tubes, in different diameters for intake and indexer, covered in white silicone tubing. The team seats the undersized silicone with the Spectrum method.
FEEDER / AMP / TRAP Indexer rollers use the same silicone-over-tube build. At the reveal the team had several Amp scoring methods and expected to change them during the season; the final Amp method is unconfirmed (see CAD). Trap unconfirmed (see CAD).	**DRIVETRAIN** Swerve, the team's 2nd competition swerve robot (OP). 8 SPARK MAX controllers run the swerve. 24x28 in frame perimeter, 75 lb with no ballast. Module type unconfirmed (see CAD).
ENDGAME Stage climb driven by 1 motor. Climb design, trap and harmony unconfirmed (see CAD).	**AUTONOMOUS** Won the Autonomous Award at FIM District Battle Creek. Routines and note count unconfirmed (not stated in the public release).
VISION A camera sits on an upright (named in a reply that the OP answered). Camera model and software unconfirmed.	**PUBLIC CAD / DESIGN** GrabCAD CAD (STEP/IGES)

LESSONS FOR STUDENTS

- Design gears for your machine: a roller-pinion profile with 3/16 in dowel pins let the team cut sector gears with one 3/16 in router bit, where their 2022 turret gear needed a 1/16 in bit and 3 operations.
- Plan inspection for custom gears: the team checked the polycarbonate gear teeth every match and the chain tensioner about once a day, and replaced the sector gear plates once in the season. The teeth cracked when the rotation encoder lost position and the motor tried to run through the lower hard stop.
- Size for speed: the OP states a small, light robot (24x28 in, 75 lb) to stay fast and avoid defense.
- Keep the control system small: 14 SPARK MAX controllers in total, with CAN utilization of 50-60% reported by the team.
- Make status LEDs serviceable: the team mounted LED strips in plastic rectangular tube with printed blocks and hot glue, had bad solder joints and connector problems at the strip base, and recommends a printed attachment instead. Rollers are white silicone tubing over polycarbonate or aluminum tubes.

GrabCAD: GrabCAD CAD (STEP/IGES) · GrabCAD CAD (STEP/IGES) · Reveal and CAD release thread (CD) · Reveal video · 2024 Champs recap video

Profile	4043 NerdHerd Notezart USA-OR	Local EPA 36 (#145) 46-16

'Notezart': first-year swerve; RI3D-style arm with flywheels and kick-up feed wheels; intake in a protective frame box

SHOOTER (SPEAKER) Flywheel shooter on an arm similar to the Unqualified Quokkas Ri3D design, with some differences. Flywheels are Stealth wheels; one side has TPU socks that add spin to the note. Motors and ratios: Unconfirmed (see CAD).	**INTAKE** The intake sits in a front 'box' frame extension that protects the arm at its lowest position. The intake does not attach directly to the drivebase. The side and front 1x2 tubes are thicker aluminum extrusions than the CAD shows, joined with custom steel L-brackets. The team chose this after testing a mock frame extension early in the season.
FEEDER / AMP / TRAP A bar of kick-up wheels on the arm feeds the note into the flywheels once they are at speed. Amp and trap mechanisms: Unconfirmed (see CAD).	**DRIVETRAIN** Swerve; 2024 was the team's first year with swerve. The drivebase follows a public offseason system: a fixed drivebase that each mechanism must fit. That offseason drivebase uses SDS Mk4i modules (team drivebase CAD release, 2024-01-06). The team flipped the swerve module brackets for more ground clearance.
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines: Unconfirmed (see code). Autonomous Award at PNW District Wilsonville.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Test the weak point early: a mock frame extension showed the team it needed custom steel L-brackets to stop skewing.
- Keep the intake off the drivebase: the intake does not attach directly to the frame, and the 'box' shields the arm at its lowest position.
- Stage the shot: kick-up wheels hold the note until the flywheels are at speed.
- Read the post with the CAD: the team states the front tubes are thicker than the CAD shows.

Onshape CAD: Onshape CAD · Reveal and CAD/code release (CD) · Onshape CAD · Code (GitHub) · Offseason drivebase CAD release (CD)

Profile	5406 Celt-X Canada	Local EPA 36 (#159) 41-21

SDS Mk4i swerve; flywheel shooter set by arm angle; roller intake; amp scoring; climber with trap mechanism

SHOOTER (SPEAKER) Flywheel shooter. The auto-shoot criteria include flywheel speed and arm angle, so an arm sets the shot angle. Before Houston the team added auto-shoot: the feeder motor runs when the flywheel is at speed, the arm is at angle, the AprilTag is centered and drive speed is below a threshold. A time-of-flight sensor sits halfway up the shooter.	**INTAKE** Roller intake behind a front cross bar. An upgraded 2x2 cross bar held up at McMaster, but a hit bent a roller 1/4 in behind it. An angled hit bent the side plates, and the intake geartrain risked derailment. Spare intake plates went on before Houston. The kickoff requirements called for a floor intake.
FEEDER / AMP / TRAP A feeder moves the note from the intake to the shooter. The team scored in the Amp; the amp mechanism type is unconfirmed (see CAD). The team reports its trap mechanism worked every time at McMaster when the drive team left time to line up and deploy it. A second time-of-flight sensor lower in the intake and a rear-facing CANdle light give the driver earlier notice of a note.	**DRIVETRAIN** SDS Mk4i swerve with CTRE CANcoders, which the team zeroes with Tuner X. Changed before Houston to NEO Vortex drive motors for more speed, with 8mm x 19mm bearings added at the top of the pinion to reduce cantilevered load on the motor shaft. 3D-printed TPU treads on SDS billet wheels; the 2024 version had solid rims that often delaminated. Gyro changed from a Nav-X to a CTRE Pigeon for the provincial championship after the Nav-X reported wrong angles.
ENDGAME Climber and trap mechanism. The kickoff requirements called for a single-robot climb, plus trap or a reliable High Note. Harmony unconfirmed (not stated in the public release).	**AUTONOMOUS** Note count unconfirmed (not stated in the public release). The kickoff target for BetaBot was a 3 to 4+ piece auto with centerline notes. At the provincial championship the team wrote a new auto to fit its partners' autos. Before Houston it added and retuned autos.
VISION AprilTag shot curves for the centered speaker tags (#4, #7) and the off-center tags (#3, #8). At McMaster a code fault made the robot use the off-center tags nearly half the time; the team disabled those curves for playoffs. A note-detection feature gives AI-assisted steering on a driver button; a pre-McMaster post says it was not yet in the autos. Camera hardware unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Test a feature against match logs: the team found its tag-selection command used the off-center AprilTag curves nearly half the time, and those shots were worse than with no feature.
- Sensor position sets driver feedback time: a second time-of-flight sensor lower in the intake signals a note earlier than the one halfway up the shooter.
- Plan where collision energy goes: the 2x2 intake cross bar held, but hits then bent a roller and the intake side plates.
- Check shaft loads before a motor swap: for NEO Vortex drive motors on Mk4i modules, the team added a bearing at the top of the pinion.
- Schedule a fallback robot: the team later wrote that AlphaBot's design fell behind, and BetaBot was done too late for much iteration.

Onshape CAD: Onshape CAD · Onshape CAD (Spectrum CAD Collection) · 2024 Open Alliance build thread (CD) · 2024 Robot Skills List (Google Sheets) · 2025 build thread with 2024 retrospective (CD)

■ Study pick

6036 Peninsula Robotics Snoopy

USA-CA

Local EPA 36 (#162)

43-19-1

'Snoopy': turreted speaker shooter on an X-contact bearing with auto-aim; custom AprilTag vision; custom trajectory app

SHOOTER (SPEAKER) Turreted speaker shooter with auto-aim; the team reports full turreted auto-aiming teleop cycles within 6 hours of build handoff. The turret rides on an X-contact bearing from Lily Bearing (NC065XP0) plus a McMaster-Carr radial bearing; the first bearing arrived about a week and a half after the quote, a later spare took about three weeks. An absolute encoder is used only to zero the turret; during the match the code uses the Kraken's internal encoder. The encoder-to-turret ratio was not a clean multiple ('I think it was 140:44?' per the OP), so the turret must start within about 1/3 of a revolution, with a dashboard readout to confirm it is not wrapped.	**INTAKE** Unconfirmed (not stated in the public release; see CAD and tech binder).
FEEDER / AMP / TRAP Unconfirmed (not stated in the public release; see CAD and tech binder).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The reveal OP posted the tag '#SwurretGang'; no post names the drivetrain.
ENDGAME Unconfirmed (not stated in the public release; see CAD).	**AUTONOMOUS** Paths come from a custom trajectory optimization app the team wrote. The captain names a fast 'pickup aim shoot pickup' 4-speaker cycle as a focus, run on sensors and conditionals instead of tuned timings. Autonomous Awards at the Arizona Valley and Hueneme Port Regionals.
VISION Custom AprilTag vision software written by the team; the captain points to the tech binder for detail. The team stress-tested its coprocessors to choose heatsinks and active cooling, and replaced camera lenses it did not like. Camera count and model are unconfirmed.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Simulation before hardware: per the software captain, the team ran full turreted auto-aiming teleop cycles within 6 hours of build handoff, mostly tuning. Bring-up rules: clamp every motor command, add software and hardware limits, and graph outputs with commands not sent while moving mechanisms by hand.
- Use sensors and conditionals instead of tuned timings: the captain states this is how the team made its 'pickup aim shoot pickup' cycle fast.
- Custom software has a cost: the team wrote its own trajectory app and AprilTag vision, and the captain states this pays off only if the team can keep it ahead of open-source tools such as Choreo or PhotonVision.
- Turret zeroing can be simple: one absolute encoder at a non-integer ratio zeroes the turret, then the motor's internal encoder takes over. The OP calls a two-encoder Chinese remainder theorem setup 'probably' the cleaner solution and would add a second encoder next time.
- Plan tolerances for outsourced parts: the turret took at least 5 or 6 mechanical iterations. The OP would split the laser-cut ring gear into two 1/8 in sheets, because the single sheet was too thick for accurate tolerances.

Onshape CAD: Onshape CAD · 2024 CAD and tech binder release (CD) · Onshape CAD · Software lessons write-up (CD) · Strategy document thread (CD) · Reveal video

■ Study pick

111 WildStang

USA-IL

Local EPA 35 (#180)

22-23

Swerve; tilting shooter with note compression; lift on the amp note path; LL3 + Coral camera steers intake alignment

SHOOTER (SPEAKER) The team built several shooter versions. The prototype had horizontal note compression and no feed system. V2 kept the compression but added feed drag; removing the top feed rollers cut slip and raised exit velocity, with less compression possible on the single-sided feed. A V3 with multiple rows of flywheels was planned for CIR; whether it was built is unconfirmed (see CAD). The shooter tilts and has a 'subwoofer' preset.	**INTAKE** A floor-facing LL3 with Coral, added for Midwest, commands the swerve strafe to line the intake up with a note. In intake mode the driver controls only throttle. The team added it after drivers had trouble aligning the intake in blind spots near the speaker and stages. Intake mechanism detail unconfirmed (see CAD).
FEEDER / AMP / TRAP Notes move from the intake note path into a separate amp note path. A lift is part of the amp scoring sequence. At CIR a failing 775 motor brush jammed the feed for 3-4 matches, and the motor failed in Q43. Changed for Midwest: one LaserCAN aimed down the feed path was too inaccurate, so the team fitted two LaserCAN sensors used as break beams. Trap mechanism unconfirmed (see CAD).	**DRIVETRAIN** Swerve. The team dropped to the EH3 ratio from the upgrade kit and runs NEO Vortex drive motors with 50A current limits. The stock REV over-molded wheels slipped and wore, so the team changed to its own TPU tread on Thrifty Bot billet hubs. Inspection weight was around 103 lbs on 2024-02-23.
ENDGAME The team tested its climb on 2024-02-25 and posted a video. Climb mechanism detail and trap capability are unconfirmed (see CAD).	**AUTONOMOUS** A 6-note center-start auto had its baseline path on 2024-03-10; the team needed to trim about 3 seconds. The note-alignment assist also runs in auto. It turns on after about 75% of each pre-planned path to correct drift. Path tool unconfirmed.
VISION At CIR the team used PhotonVision on an Orange Pi 5 with Arducam 844 sensors; camera pitch moved with the shooter angle. The team reported exposure 'hunting' and hardware stability problems. Changed for Midwest to two crossed Limelights on top of the lift static stage. From the stage range, the edges of their wide view can see the Amp AprilTags past blocker robots.	**PUBLIC CAD / DESIGN** 2024 CAD (Onshape)

LESSONS FOR STUDENTS

- Shooter exit velocity depends on contact time and slip: the team traced a V2 loss to drag from the feed rollers.
- A camera can do the fine alignment: the floor-facing LL3 with Coral controls strafe while the driver controls throttle.
- Sensor layout matters: one LaserCAN measuring note distance was too inaccurate, and two break-beam LaserCANs replaced it.
- Mount localization cameras on fixed structure: moving them off the tilting shooter removed a camera-pitch calculation.
- Event reliability: the CIR review lists a failed feed motor, a loose battery that destroyed a roboRIO, and a power wire pulled from the PDH.

Onshape CAD: 2024 CAD (Onshape) · 2024 CAD (Onshape) · 2024 build blog, teams 111 and 112 (CD) · 2024 climb test video (YouTube)

■ Study pick

2877 LigerBots

USA-MA

Local EPA 35 (#192)

35-27-1

Under-bumper intake; shooter pivots on an angled cascade elevator; winch climber; MK4i swerve; split-height bumpers

SHOOTER (SPEAKER) The 2024-01-31 CAD overview shows a pivoting shooter on an angled elevator. The pivot is a NEO with an overall ratio of 22.5:1. The team had two shooter designs, planned to prototype both, and made the CAD configurable to either. Side/horizontal design: 4 sets of two 2.875 in BaneBots wheels, one NEO Vortex direct-driving each side. Top/bottom design: dimensions taken from 6328's shooter, 4 sets of three 2.875 in BaneBots wheels on cantilevered hex shafts, 2 NEO Vortex motors with a 1:2 increase. The posts do not state which design was used (Unconfirmed, see CAD).	**INTAKE** Under-bumper intake. One roller of 1.375 in BaneBots wheels kicks up the note. Then come 3 polycarbonate rollers covered in silicone tubing and 1 roller of 2 in flex wheels. The team reported good intake test results on 2024-02-09. Intake motor unconfirmed (see CAD).
FEEDER / AMP / TRAP Feeder: a NEO drives 2 sets of 2 in flex wheels through a 2:1 reduction, with polycord to reverse direction on the opposite side. Angled elevator: 1 NEO at 4:1, cascade rigging with belt on the first stage and rope on the second. It uses AM bearing blocks, 2x1 WCP punched tube and custom CNC gussets. Changed in March: the elevator was rebuilt with upgraded brackets and cross bars. Amp and trap method unconfirmed (see CAD).	**DRIVETRAIN** Swerve with MK4i modules at L2 gearing; the team chose L2 'because that is what we have'. NEO drive and NEO steer as of 2024-01-31; the team said it may switch to Falcons. The final drive motor is unconfirmed. 23 in x 27 in drive base, 29.25 in x 27.5 in frame perimeter. Steel box tube frame rails to resist bending and lower the CG. Split bumpers: the front is low to push notes away, the back is high so the note passes under.
ENDGAME Winch climber, listed as work in progress on 2024-01-31. Two independent NEOs, each with a 15:1 reduction, one per side. The low front bumper gives the chain more room to sit low in the robot. Trap and harmony unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Bumper height can be a design variable: a high back bumper lets the note reach the under-bumper intake, and a low front bumper gives the stage chain room.
- One configurable Onshape model held two shooter concepts while the team prototyped both.
- The elevator post states motor, ratio, rigging and structure in one list; check each item against the CAD.
- The team states a reason for its frame material: steel box tube rails to resist bending and to lower the center of gravity.

Onshape CAD: Onshape CAD · Onshape CAD (Spectrum) · 2024 Open Alliance build thread (CD)

■ Study pick

9312 NERD Spark BATMAN

USA-MI

Local EPA 35 (#187)

33-23

'BATMAN': swerve; intake-shooter tilted by a printed CF 4-bar; 2 DOF arm with wrist for amp and chain climb

SHOOTER (SPEAKER) Prototype (2024-01-23): four 4 in Thrifty solid urethane wheels (45A) on two NEO Vortex at 1:1 direct drive, open loop at 100% output, with 1/2 in compression. The wheels touch only one side of the note to spin it, at a cost in forward speed. Shot angle about 35 deg, 5-6 in off the floor; the team then began implementing closed-loop velocity PID. On the competition robot a 4-bar tilts the combined intake-shooter; the team wanted higher shots over its back side when close. In a 2024-03-05 post the OP believes the team limits its NEO Vortex motors to 40 amps.	**INTAKE** Combined ground intake and shooter, planned to use about 60% of the chassis. Every intake part is Nylon 6 CF. One connecting member of the intake-shooter is designed as the weakest part so it breaks first. The team expects the aluminum hex shafts to fail on a direct hit and planned to cut spares. Motors unconfirmed (see CAD).
FEEDER / AMP / TRAP Amp: the shooter passes the note into a passive capture mechanism on a 2 DOF arm. A wrist motor with the team's NEO 550 harmonic gearbox sets the note flat against the amp wall (programmed 2024-02-26). At Jackson the transfer relied on gravity to feed the note down; the team reports it was slow and hard to line up. Trap mechanism unconfirmed.	**DRIVETRAIN** Swerve. The 24 in wide by 28 in long chassis was generated from the team's reconfigurable Onshape swerve chassis model; a sponsor laser cut the parts from the exported DXFs. The swerve pods use Thrifty Absolute Magnetic Encoders. Module type unconfirmed (see CAD).
ENDGAME Chain climb with the arms. Hooks on the forearm drive belt pass the chain to hooks on the back arm, which pull it down to latches. At Jackson the chain hung about 1.5 in higher than the team's practice chain. The team tried to climb in the first 3 matches and stopped, as the arms would break before the climb finished.	**AUTONOMOUS** At Jackson one shooting map served both alliance colors in auto and teleop; tuning it for one case broke the other three. The team split red and blue autos and used fixed set points for the 4-bar. By the end of qualifications the robot scored 3 of 4 notes in auto on both colors, and all 4 in the last qualification match, per the team. Path tool unconfirmed.
VISION A shooting map uses robot pose from vision and odometry to turn the robot and set the 4-bar tilt. The team reports vision and pose estimation issues in teleop at Jackson. Camera and software unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** 2024 CAD (Onshape)

LESSONS FOR STUDENTS

- A parametric chassis saves build time: the team set 24 x 28 in, exported DXFs from its reconfigurable Onshape model, and had laser-cut parts back in under a week.
- Tune and test a pose-based shooting map per alliance color and per mode. The team's single map broke for three cases when tuned for one, and set points fixed the autos.
- Check climb reach against field tolerance: a 1.5 in difference in chain height between the practice field and the event stopped the climb.
- Plan where the robot breaks: the team made one intake-shooter member the weakest part and planned to cut spare hex shafts before its first event.
- Measure cycle time on a new mechanism early. The gravity-fed amp transfer worked but was slow; the team writes that 'cool and interesting does not equal effective'.

Onshape CAD: 2024 CAD (Onshape) · 2024 build thread (CD) · 2024 CAD (Onshape) · Amp wrist harmonic gearbox CAD (Onshape) · NEO 550 harmonic gearbox thread (CD, 2023 offseason) · Shooter prototype video · Robot video, 2024-02-22 · Robot release video, 2024-03-29

■ Study pick

8033 Highlander Robotics Banshee

USA-CA

Local EPA 34 (#208)

33-13

'Banshee': pivoting shooter on a Maxspline shaft, 2 independent wheel sets; cascade elevator amp mech; UTB intake; Mk4i

SHOOTER (SPEAKER) Two independently controlled sets of Stealth wheels create spin. A Maxspline cross shaft takes virtually all the torsional load, so a chain on one side drives the shooter without twisting or flexing. The same Maxspline is the pivot, riding on a .75 diameter extrusion with needle bearings. A dead-axle roller for the notes rides on top of it through a large thin-section bearing from TTB.	**INTAKE** Under-the-bumper intake with 2 sets of dead-axle rollers; the front one is wrapped in silicone. Two sets of powered 4 in compliant wheels center the note. Pulleys with oversized flanges are used throughout, and the team recommends them on long belt runs such as the centering wheel assembly.
FEEDER / AMP / TRAP Cascade elevator rigged with cord (written 'UHW' in the post) and tensioned with WCP ratchet plates. Two Krakens on MAXPlanetary gearboxes drive it through chain with TTB chain combs. The amp mechanism uses several sets of rollers to deposit the note and hold it laterally. The original geometry could score in the trap with arms, but the team reports this was not reliable.	**DRIVETRAIN** Mk4i modules with Kraken X60s. The team started at L3.5 and moved to L2.5 to push through defense on its 3D printed TPU wheels. The wheel CAD is public; team replies give the print setup (Overture TPU 90a, 0.6 mm nozzle, about 8.5 hours per wheel, about 3 per tournament) and Onyx hubs at about 40% infill.
ENDGAME A servo-actuated latch on the elevator is used for the climb. The trap arms on the amp mechanism carry hooks for a backup climb, which the team reports worked well. Trap scoring was not reliable.	**AUTONOMOUS** Unconfirmed (not stated in the public release; see code).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- One shaft, three jobs: the Maxspline cross shaft is the torsion member, the shooter pivot and the support for a note roller.
- A shaft that takes the torsional load lets a wide shooter be chain-driven from one side without twist.
- Gear the drivetrain for the game: the team moved from L3.5 to L2.5 to push through defense.
- Use pulleys with oversized flanges on long belt runs, as on the centering wheel assembly.
- Robustness has a weight cost: the team reports 124 lb at the Championship and names it as something to avoid in the future.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Code (GitHub) · TPU drive wheel CAD (Onshape)

■ Study pick

341 Miss Daisy Miss Daisy XXII

USA-PA

Local EPA 33 (#247)

38-22-1

'Miss Daisy XXII': full-width over-bumper intake feeding an under-bumper intake; belts on the swerve modules center notes

SHOOTER (SPEAKER) Unconfirmed (see CAD). The release thread does not describe the shooter.	**INTAKE** Full-width over-the-bumper intake that can grab notes at the extreme edges of the robot. An under-the-bumper intake then takes the note. Timing belts wrapped around the intake-side swerve modules pull edge notes to the center before they reach the under-the-bumper intake.
FEEDER / AMP / TRAP The note passes from the intake into a conveyor, per the OP's description of the intake test. Amp and trap mechanisms unconfirmed (see CAD).	**DRIVETRAIN** Swerve, per the OP's intake description. Module type and ratio unconfirmed (see CAD). The team chose belts around the modules over a non-standard drivetrain to center notes.
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release; see code).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Two intakes in series: a full-width over-the-bumper stage for reach, then an under-the-bumper stage.
- The centering belts ride around the swerve modules, so the team kept a standard drivetrain and still centered notes picked up at the robot's edges.
- The release pairs CAD with code and a design binder for the same robot, so a student can check one against the others.

Onshape CAD: Onshape CAD · CAD, code and binder release (CD) · Onshape CAD · Code (GitHub) · 2024 design binder (Google Drive)

■ Study pick

2930 Sonic Squirrels Maestro

USA-WA

Local EPA 33 (#238)

29-26-1

'Maestro': continuous belt-driven elevator; PhotonVision note detection and localization; Choreo autos

SHOOTER (SPEAKER) Unconfirmed (not stated in the public release). Read the shooter design from the Onshape CAD.	**INTAKE** Unconfirmed (not stated in the public release). Read the intake design from the Onshape CAD.
FEEDER / AMP / TRAP The release describes a continuous, belt-driven elevator. The team says 2910's 2018 robot inspired it. The belt runs are outside the tube, not inside, for maintenance, repair and making spares. The code has teleop auto-align to both the amp and the trap. The feeder and the amp and trap end effector are unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The release states the robot uses 1678-style bumper mounts for quick bumper swaps.
ENDGAME The code has teleop auto-align to the trap, per the release post. The climb mechanism and harmony are not described; unconfirmed (see CAD).	**AUTONOMOUS** Trajectories come from Choreo. A state machine lets the robot make decisions during auto, for example whether to skip a game piece. AdvantageKit handles replay, logging and simulation. The team won the Autonomous Award at the PNW District Championship. The note count of each routine is unconfirmed.
VISION PhotonVision for pose localization. A PhotonVision neural network detects game pieces, and the code uses it to adjust paths on the fly. Camera count and camera models are unconfirmed (see CAD).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Use CAD configurations to communicate intent: the top-level model has one configuration per scoring setpoint and an 'adjustable' configuration for the full range of motion.
- The continuous belt-driven elevator puts the belt runs outside the tube. The team chose this for maintenance, repair and making spares.
- The team names a state machine that can skip a game piece in auto, PhotonVision game piece detection and Choreo trajectories as the software work behind its Autonomous Award at the PNW District Championship.
- The team adopted 1678-style bumper mounts for quick bumper swaps.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Reveal thread (CD) · Reveal video

■ Study pick

930 Mukwonago BEARs

USA-WI

Local EPA 32 (#290)

35-16-1

Turreted shooter driven through a strain-wave gearbox; 3D printed gears; SDS MK4i L3 swerve with Krakens; about 113 lb

SHOOTER (SPEAKER) Turreted shooter. The turret is driven through a strain-wave (harmonic) gearbox, sourced by a team mentor; the OP reports zero backlash, no noticeable wear all season, and a smaller package. A hex shaft through the turret carries the turret encoder. Shooter gears are 3D printed; the ratio changed from 2:1 to 1.5:1 after the Wisconsin Regional by increasing gear DP without changing the plates.	**INTAKE** The intake bevel gears are 3D printed; the OP reports no issues with them. Roller layout and motors unconfirmed (see CAD).
FEEDER / AMP / TRAP Unconfirmed (see CAD).	**DRIVETRAIN** SDS MK4i swerve modules, L3, with Krakens and 16t motor pinions to raise top speed. The OP states this is likely the main factor in the robot's speed. Weight about 113 lb at the Championship, and lighter at Lake Superior.
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release; see code).
VISION Unconfirmed (not stated in the public release). The OP mentions 3D printed camera mounts.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A strain-wave gearbox can drive a turret in a small package; the OP reports zero backlash and no noticeable wear over the season.
- A hex shaft on the turret axis gave a direct mount for the encoder, without the separate encoder gearbox the team built in 2020/21.
- Printed gear practice from the thread: the OP recommends about 10 DP, and the team printed with 4 walls and gyroid infill. The OP names wall count as probably the most impactful setting, because the bore is the main failure point.
- Printed gears can absorb a ratio change: the shooter went from 2:1 to 1.5:1 after the Wisconsin Regional with new gears on the same plates.
- The failure is documented too: the bore of the motor-driven shooter gear rounded out twice, once with the Kraken shaft profile printed in and once with a hex insert.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video

■ Study pick

2102 Team Paradox

USA-CA

Local EPA 32 (#262)

19-10

Combined intake/shooter with two roller pairs (shoots from either side); climber; swerve; REV motors only

SHOOTER (SPEAKER) Combined intake/shooter mechanism. Software stows the note in the correct pair of rollers and spins up the other pair, depending on the side the robot shoots from. In speaker mode the robot aims passively to shoot or pass while it holds a note. The team states it uses REV motors only; motor models and ratios are unconfirmed (see CAD).	**INTAKE** Part of the combined intake/shooter. The intake retracts automatically when a note is acquired. A 12-pin Deutsch connector and an Anderson block connect the chassis to the pivot, so the mechanism can be hot-swapped. Roller details unconfirmed (see CAD).
FEEDER / AMP / TRAP No separate indexer is described; the note is stowed in the rollers of the combined intake/shooter. An operator switch toggles between amp and speaker modes. Amp and trap hardware unconfirmed (see CAD).	**DRIVETRAIN** Swerve, per the team's 2024 wiring post, which shows the swerve module CAN connectors. All wires of one type for a module share one connector, so motors cannot be plugged in wrong during a module swap. Module type and ratio unconfirmed (see CAD).
ENDGAME A climber, run from one operator button. The climber has its own master disconnect point for hot-swapping. Trap and harmony unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Team-written AprilTag detection software, released on GitHub as the software used in the 2024 season. Camera hardware unconfirmed.	**PUBLIC CAD / DESIGN** 2024 robot CAD (Onshape)

LESSONS FOR STUDENTS

- Trunk/branch CAN: each CAN device sits on its own short branch, so one bad connection does not take down the whole bus. The team states FIRST does not recommend this layout and that stub lengths must stay short.
- Plan wire paths in CAD: the team sketches wire routes and zip-tie holes in Onshape before wiring anything.
- Master disconnects: all wires for a hot-swappable mechanism pass through one or two connectors, here a 12-pin Deutsch connector and an Anderson block between chassis and pivot.
- Automation to cut driver load: by the team's own estimate, about 80% of matches ran on 4 buttons besides the drive sticks. The team also states the code was hard to debug and does not recommend it as a reference.
- Prototype early: a reused prototyping robot let the team run full cycles by the Wednesday after kickoff and test the intake/shooter combination.

Onshape CAD: 2024 robot CAD (Onshape) · Open Alliance thread with 2024 write-ups (CD) · 2024 robot CAD (Onshape) · 2024 code (GitHub) · AprilTag software (GitHub) · Reveal thread (CD)

Profile	7312 T3 USA-TX	Local EPA 32 (#293) 37-22

Fixed-angle two-sided shooter with rotary amp bar, OTB rotary intake, GreyT telescoping climber; Mk4i swerve (Jan 2024 design)

SHOOTER (SPEAKER) Two-sided shooter with ION grip wheels, driven by two separate NEOs at 1.5:1. The note is compressed to 7 in wide. Fixed angle, per the selected Concept 2. Values are from the 2024-01-19 design update.	**INTAKE** Over-the-bumper rotary intake, changed from an under-the-bumper design in week 2. Sets of 2 in Thrifty squish wheels are driven by a Falcon (the post says 'maybe Kraken'), and the intake pivots on a NEO geared 40:1. Three hex shafts near the end of the intake add rigidity and guide the note.
FEEDER / AMP / TRAP Amp: a rotary bar works with the fixed-angle shooter. The team's testing found an upward shot stayed in the amp when a piece of box tube sat about halfway up the opening. Trap was not pursued; the team judged it unattainable. Indexer detail unconfirmed (see CAD).	**DRIVETRAIN** Mk4i swerve, listed as a common part of both January concepts. Drive motors and ratios unconfirmed (see CAD).
ENDGAME Stage climb on GreyT telescoping tubes at 17.5:1. The hook design was still in progress on 2024-01-19. The team ranked ONSTAGE above the amp because the ranking point needs two robots ONSTAGE. Trap not pursued.	**AUTONOMOUS** Unconfirmed (not stated in the public release)
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Full robot assembly (Onshape)

LESSONS FOR STUDENTS

- A week 1 priority list: speaker, then ONSTAGE, then amp, with trap dropped as unattainable.
- A fixed-angle shooter plus a rotary bar can score the amp; the team chose this over a pivoting shooter to finish before a week 1 event.
- Shooter numbers to compare against other teams: two NEOs at 1.5:1 on ION grip wheels, note compressed to 7 in.
- The intake moved from under the bumper to over the bumper during week 2 of design; the CAD shows the result.

Onshape CAD: Full robot assembly (Onshape) · 2024 build thread (CD) · Full robot assembly (Onshape) · Mechanisms CAD (Onshape)

Profile	1710 The Ravonics Revolution ADONIS USA-KS	Local EPA 31 (#316) 29-13

'ADONIS': swerve robot with an intake, wrist, shooter and a rope-rigged elevator; detail in the Fusion 360 CAD

SHOOTER (SPEAKER) The 2024 robot has a wrist and a shooter; the team's preseason programming lessons use both. At the Daly Division the team fed notes across the field to alliance partners; its match review states it was not confident passing from the middle of the field. Shooter type, motors and ratios are unconfirmed (see CAD).	**INTAKE** The intake frame uses 1x1 in steel tubing bolted to the robot frame, with a Delrin block locking system. At Cow Town Throwdown the tubing bent, the bolt joining the intake bar to the frame sheared, and the Delrin lock failed; the team added rivets to hold the intake. The team's programming lesson for this intake uses beam breaks to sense the note.
FEEDER / AMP / TRAP An elevator rigged with ropes on the center and left side. The team reports rope fraying and snapping all season; at Cow Town Throwdown the left rope snapped and the elevator did not work for the playoffs. A polycarbonate hard stop also broke. The elevator's scoring role (Amp, Trap) is unconfirmed (see CAD).	**DRIVETRAIN** Swerve. The Cow Town Throwdown post names the rear 2x1 in swerve tubing. After the season the team kept the ADONIS chassis as a testbed named ALPHA. The team states it decided to change to L3 gearing; the chassis has Kraken drive and Falcon steer motors. The 2024 module type and ratio are unconfirmed (see CAD).
ENDGAME Stage chain climb. The team's Daly Division match review describes the robot climbing at center stage in Qualification Match 120. Trap and harmony are unconfirmed (not stated in the public release).	**AUTONOMOUS** 4-note auto that scores the close notes. In Daly Division Qualification Match 120 the robot used note detection to pick up a note passed by 118 during auto and score it in the speaker. A preseason programming lesson used PathPlanner; the path tool for the competition autos is unconfirmed.
VISION PhotonVision. The team's programming lead lists an AprilTag camera and a note-detection camera on the 2024 robot. Camera models are unconfirmed.	**PUBLIC CAD / DESIGN** Fusion 360 CAD

LESSONS FOR STUDENTS

- Log failures by subsystem after each event. The Cow Town Throwdown post lists each failure and whether the team fixed it or left it as not critical.
- Note detection can earn points in auto beyond the planned path. In a Daly Division match 1710's robot detected a note passed by an alliance partner, picked it up and scored it during its 4-note auto.
- Rope rigging on an elevator is a wear item. 1710 reports frayed and snapped ropes all season, and one snap took the elevator out for a full playoff.
- A retired competition chassis can serve as a software testbed. The team kept the ADONIS swerve base as ALPHA to test a new AdvantageKit swerve template before the 2025 kickoff.
- The team states its recent robots weighed 120-125 pounds without bumpers and battery, and set a lighter target for 2025.

Fusion 360 CAD: Fusion 360 CAD · Fusion 360 CAD (2024 competition robot) · 2024 Open Alliance build thread (CD) · 2025 build thread with 2024 robot offseason notes (CD) · Scouting data, Daly Division (team site)

■ Study pick

1155 SciBorgs Taiwanese Fried Chicken

USA-NY

Local EPA 30 (#334)

18-9

'Taiwanese Fried Chicken': pivoting flywheel shooter that also scores Amp; shooting on the move from pose estimation; swerve

SHOOTER (SPEAKER) Flywheel shooter on a pivot. The team re-tuned the pivot and shooter PID after NYC. The team simulated shot trajectories during development and wrote a public shooting-on-the-move command. Flywheel motors and ratios unconfirmed (see CAD).	**INTAKE** Roller intake with beam-break sensors. The team moved the beam breaks from soldered joints to Molex SL connectors; a swap then took about 3 minutes instead of about 30. An intake roller shaft failed after a hard hit at RoboReplay and was replaced.
FEEDER / AMP / TRAP A conveyor feeds the shooter. Amp scoring runs through the shooter; 3D-printed shields around the flywheels were added to prevent note jams while amping. The team reports an inconsistent Amp angle at Duel on the Delaware. A trap mechanism is not described; unconfirmed (see CAD).	**DRIVETRAIN** Swerve: the team's SOTM talk covers shooting on the move with a swerve drive. Module type unconfirmed (see CAD).
ENDGAME Stage chain climb with climb-lock hooks and climb plates. One hook broke in a RoboReplay qualification match and the robot still climbed. Trap and harmony unconfirmed (see CAD).	**AUTONOMOUS** A center 5-note auto ran at RoboReplay, where the team won the Autonomous Award. The team says its autos were weaker earlier in the season. Path tool unconfirmed (see code).
VISION Cameras feed pose estimation used for shooting. The team's running theory is that a slowly loosening camera mount degraded pose estimation and shot accuracy. It found and fixed the mount at RoboReplay. Camera model and vision software unconfirmed; the robot also carries an Orange Pi whose role is not stated.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Check sensor mounts when accuracy drifts: the team first suspected pivot slop, then found a loosening camera mount that it believes caused falling shot accuracy.
- Put connectors on sensors that get hit: moving the intake beam breaks to Molex SL connectors cut a swap from about 30 minutes to about 3.
- SOTM is documented three ways: a build-thread post on shot simulation, a talk, and the public Shooting.java command.
- Guard the places where notes jam: the team added 3D-printed shields around the flywheels to prevent note jams when scoring in the Amp.

Onshape CAD: Onshape CAD · Onshape CAD #1 · Onshape CAD #2 · Shooting command code (GitHub) · 2024 build thread, SOTM post (CD) · Open Alliance thread with 2024 off-season notes (CD)

Profile	3136 O.R.C.A. JT USA-VA	Local EPA 30 (#367) 38-23-1

Arm-mounted intake that shoots notes out its back with two flywheels; AprilTag distance sets arm angle and speed; 24x24 in frame

SHOOTER (SPEAKER)
Per the build log, the intake is also the shooter: it mounts on arms, takes notes in with 4 sets of wheels at the front, and shoots them out the back with two flywheels for both Speaker and Amp. After compression tests the team chose 4 in green compliant wheels. In the first build weekend the team built V1 and V2 of the intake and designed V3. Changed in late March 2024: the log refers to a new shooter with more power, aimed by AprilTag distance. The team reports accurate shots from near the Speaker to about mid-Stage, falling off past the wing. Shooting on the move was tried and dropped for time. A lob shot from outside the opponent's wing to the team's amp area fed notes to partners.

INTAKE
Wide intake with 4 sets of wheels at the front; notes pass through to the flywheels at the back. Early tests found 1 in of compression works well for notes. Changed in late March 2024: the team redesigned and built a new intake after prototyping. At the District Championship a hit cracked the intake; the team patched it with washers and polycarbonate scraps for the rest of the playoffs. Motors unconfirmed (see CAD).

FEEDER / AMP / TRAP
An arm carries the intake and shooter. The arm attachment used a temporary sprocket of about 66T, with a planned change to 68T. In late March 2024 the arm moved to a trapezoidal motion profile for smoother, more accurate aiming. Amp scoring uses the same mechanism; the Trap was not a design goal.

DRIVETRAIN
24x24 in frame. The build log mentions rebuilding swerve modules during build season. Module type unconfirmed (see CAD).

ENDGAME
Chain climb was a kickoff goal with no design chosen at the time; the Trap was not a design goal. Climb hardware unconfirmed (see CAD).

AUTONOMOUS
Kickoff goals were a Speaker shot and a taxi in auto, with no Amp in auto. Students began learning PathPlanner during build. Before the District Championship the team ran a 2 note center auto and a 3 note auto that shoot from the pickup spot. Longer paths drifted left past mid-field, so the team switched to Choreo with help from team 1599. The final autos were an amp-side 3 note and a center 4 note. Won the Autonomous Award at the CHS District Glen Allen VA Event.

VISION
Limelight cameras (the team's posts say 'limelights'). Added in late March 2024: the distance to the AprilTags sets the arm angle and shooter speed. FMS issues blocked the Limelights in the team's first District Championship match.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Sketch a concept in CAD before you build it: the team dropped a double-wristed arm after the CAD sketch showed it was not viable.
- Learn from Ri3D: the chosen layout is close to the Penn State Ri3D robot but shoots out the back of the intake, and the team found it matched the Unqualified Quokkas Ri3D design.
- Test compression early: tests led to about 1 in of intake compression and 4 in green compliant shooter wheels.
- A long-range lob can be a role: the team says it was picked at the District Championship for lobbing notes to its partners.
- Secure the roboRIO and its power wire: a loose roboRIO and a loose power wire end shut the robot down in both District Championship finals matches.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · 2024 Open Alliance build log (CD) · Build logs (Google Drive)

■ Study pick

3729 The Raiders Mr. Mimlitz

USA-CO

Local EPA 30 (#328)

15-13-1

'Mr. Mimlitz': swerve; hybrid under-bumper intake with slapdown and printed funnels; speaker, amp, Stage hang; 5-note auto

SHOOTER (SPEAKER) Scores in the SPEAKER, per the release post. Shooter type, motors and ratios: Unconfirmed (see CAD).	**INTAKE** Hybrid intake: a 10.5 in wide under-the-bumper intake, a 'slapdown' that pulls notes in over the bumper, and two passive PETG funnels below the robot to center the note. Funnels were printed at 100% infill on Bambu X1Cs. Slapdown moving plates began as 1/8 in aluminum and changed to 1/4 in Delrin after a collision at a scrimmage. Static plates bent at Central Missouri and changed from 1/8 in to 1/4 in. A Delrin plate broke in CODE Q40; the pit crew repaired it before the next match.
FEEDER / AMP / TRAP The robot places notes in the AMP, per the release post. Mechanism: Unconfirmed (see CAD). Trap: Unconfirmed.	**DRIVETRAIN** Swerve, per the release thread. The frame perimeter is 28 in on the intaking side; swerve modules and cross-rails take 17.5 in of that. Module type: Unconfirmed (see CAD).
ENDGAME Hangs on the STAGE, per the release post. Trap and harmony: Unconfirmed.	**AUTONOMOUS** 5-note auto, per the release post. The team chose a wide pickup partly for a faster center-line auto, because the robot travels less to reach each note. Path tool: Unconfirmed (see code).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Budget the frame width first: swerve and cross-rails took 17.5 in of a 28 in side, which left 10.5 in for the intake.
- Tie intake width to strategy: the team wanted a wider pickup for driver visibility and a faster center-line auto.
- Plate material follows failures: 1/8 in aluminum moving plates became 1/4 in Delrin, and 1/8 in static plates became 1/4 in.
- A helper can fail without stopping the robot: after a slapdown plate broke in CODE Q40, the robot still picked up notes for the rest of the match.
- Write the retrospective: the team states a wider 95-style under-bumper intake, with two fewer motors, would have suited it better.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Reveal (CD)

Profile	5431 Titan Robotics Orpheus USA-TX	Local EPA 30 (#341) 22-23

'Orpheus': swerve; V2 rack-and-pinion shooter angler (about 35-55 deg); pivot intake scores Amp; chain-driven Vortex climber

SHOOTER (SPEAKER)
V1 was a static shooter set at 55 deg, which prevented Amp and Trap scoring. V2 adds an 'angler': a curved rack and pinion, about concentric to the intake pivot, moves the shooter about 20 deg, from about 35 to about 55 deg relative to the note path. The shooter rides on flangeless bearings on eccentric hex shafts that run along a curved 'rib' track; the team went through three iterations of this layout. In January tests the bottom wheels needed 20 to 30% less power than the top wheels to reach the Speaker. A January prototype design whose wheels tore into the note was scrapped; the post does not say if it was an intake or shooter prototype. On 2024-03-09 the team replaced the worst shooter wheels and planned to replace a churro shaft with hex shafts. A revised shooter design was in progress on 2024-03-13.

INTAKE
Pivoting intake, part of an intake-shooter combo. Rollers are polycarbonate tubes wrapped in silicone tubing, with a MAXPlanetary gearbox and small polycord pulleys. V2 uses two NEO 550s instead of one; the team states this made Amp scoring more consistent without overloading the single V1 NEO 550. V2 adds beam break sensors, a protective panel and bearings that let the intake glide on the carpet. The V2 pivot moves the pulleys and belts inside the base plates, because the swerve modules blocked repairs on V1. This stage gives a 2:1 torque reduction. In January, after REV recalled their motor controllers, the backup plan was a NEO for the pivot and a NEO 550 for the rollers; the reduction changed from 15:1 to 5:1 and the NEO 550 then would not run.

FEEDER / AMP / TRAP
The Amp is scored with the intake: the 2024-03-04 practice notes (posted 2024-03-29) describe an auto command that outtakes the intake to score in the Amp. The V2 write-up says the static V1 shooter prevented Amp and Trap scoring, and introduces the angler. The drive team practiced Amp and Trap scoring on 2024-02-21. A separate feeder and the Trap method are unconfirmed (see CAD).

DRIVETRAIN
Swerve with a 29x29 in frame perimeter (V1 plan). On V2, cross beams set the spacing of the intake base plates and shooter plates and stiffen a belly pan drilled with nearly 300 holes on a 1 in pattern for electronics on its underside. A polycarbonate skid plate covers the electronics. The team used CTRE swerve code with Phoenix 6, traced a clicking sound in the modules to a bearing problem, and almost fixed a math issue that stopped the robot at 45 deg joystick positions. Module type is unconfirmed (see CAD).

ENDGAME
The V1 plan was an elevator climber. The V2 climber is chain driven by NEO Vortex motors: a chain bolted to the traveling arm runs between a top sprocket and a drive sprocket on a 1/2 in hex shaft. Bearing blocks act as hard stops, and the team rejected an earlier pulley design as too fragile. Retracted, the robot fits under the Stage and chain. On 2024-03-06 the climber ratio changed from 27:1 to 9:1 and pegs were added to the hooks. The right arm's chain was loose and skipped, the climber broke on 2024-03-09, and on 2024-03-14 the team removed an arm while a new climber was designed. Trap and harmony are unconfirmed.

AUTONOMOUS
On 2024-03-06 the team confirmed a 1-piece preload Amp auto. A PathPlanner path ending interrupted the Amp outtake command; the mentor's fix was to put the event in the auto, not in the path. The code uses PathPlanner 2024.1.2. Other routines unconfirmed.

VISION
The team calculated a Limelight 3 mount angle of 45 deg, up from a first estimate of 22 deg, for close-range AprilTag accuracy. They also calibrated PhotonVision. The leads concluded a coprocessor would help with more cameras and that a standalone camera might replace the Limelight. In February the team confirmed an Orange Pi worked but was laggy, and suspected the ethernet cables. Final setup unconfirmed.

PUBLIC CAD / DESIGN
Fusion 360 CAD (V2)

LESSONS FOR STUDENTS

- Add a degree of freedom when a fixed design blocks a task: the static 55 deg V1 shooter could not score the Amp or Trap, so V2 added a 35-55 deg rack-and-pinion angler.
- Design for repair: V1 intake pivot belts sat outside the base plates where the swerve modules blocked access, so V2 moved them inside.
- Shot shape can be tuned with wheel speed: 5431 found the bottom wheels needed 20 to 30% less power than the top wheels to reach the Speaker.
- Calculate a camera mount angle before you build. The team's calculations changed the Limelight angle from a first estimate of 22 deg to 45 deg for close-range accuracy.
- Plan for supply problems. When REV recalled their motor controllers, the team made a backup plan with a NEO and a NEO 550 on the intake.

Fusion 360 CAD: Fusion 360 CAD (V2) · Fusion 360 CAD (V2) · Fusion 360 CAD (V1) · 2024 Open Alliance thread (CD)

■ Study pick

1414 IHOT Encore

USA-GA

Local EPA 29 (#420)

41-21

'Encore': Orbit-inspired layout; differential speed shooter; amp deflector; combined intake/indexer; REV swerve; no climb

SHOOTER (SPEAKER) Differential speed shooter, per the release post. The shooter pivots. Before the anti-jammer, the team states the robot could intake from 0 deg (shooter flat down) up to 70 deg, and moved past 70 deg only to score in the amp. Motors and ratios: Unconfirmed (see CAD).	**INTAKE** The intake and indexer are one combined system. Rollers are mostly dead axle: the axle slides through the side plate, so the plate needs no bearing. Roller stock is 1.125 in ID x 1.25 in OD with 0.0625 in silicone tubing. Pulleys are 34t 3mm GT2 with 0.500 in ID x 0.875 in OD radial bearings (WCP-0777) inside.
FEEDER / AMP / TRAP An 'amp deflector' moves into the path of the note to score in the amp without the precise controls Orbit used. For Championship the team added an 'anti-jammer': two plates with a roller and belts. Before it, scoring in the amp with a note lying on the floor in front of the robot caused a jam. With it, the robot can intake at any shooter angle. Trap: none.	**DRIVETRAIN** Swerve with REV swerve modules, per the OP. The OP states the REV modules were the only mechanical failure of the season.
ENDGAME No climb and no trap. The OP states both were omitted mainly due to time, because the design started after Orbit's release video.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Copying a published layout still leaves decisions: the team replaced live-axle rollers and bearing plates with dead axles that slide through the plate, to match what it could machine.
- A moving deflector can replace control precision: the 'amp deflector' moves into the note path so amp scoring does not depend on exact shooter control.
- Shared intake and indexer paths create interactions: amping with a note lying on the floor caused a jam until the team added an 'anti-jammer' roller.
- Scope cuts are a design choice: the team omitted climb and trap for time after starting a late redesign.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD

■ Study pick

6647 PrepaTec - VOLTEC

Mexico

Local EPA 29 (#415)

34-21-1

Swerve NOTE robot; elevator removed mid-season; Kraken shooter and intake; Limelight 3 note detection; amp auto

SHOOTER (SPEAKER) The team prototyped shooter wheels of several types for compression and grip on the note. At Monterrey the shooter sat on top of the elevator; the team states this affected its shots. Shooter motors changed from NEO to Kraken during the season. The pivot used gears; after Hermosillo the team planned a chain pivot to cut backlash and a change to add spin.	**INTAKE** The team describes a touch-it-own-it intake that picks notes from different angles. To close the intake, the code applies equal output to both of its motors in the same direction. Intake motors changed from NEO to Kraken during the season. Note detection used current limits and a debouncer at first; changed at Monterrey to a 50cm beam break.
FEEDER / AMP / TRAP Feeder and amp mechanism unconfirmed (see CAD). The robot scored in the amp: at Hermosillo it placed two amp notes, then one speaker note while amplified. After Hermosillo the team planned to test the intake as the storage mechanism. Trap mechanism unconfirmed; the team lists trap scoring as not always needed.	**DRIVETRAIN** Swerve, per the build thread. Drive motors were Falcon 500 at the start of the season. Changed before Hermosillo to Kraken motors. Module type unconfirmed (see CAD). The swerve code is the team's AndromedaLib library, moved to AdvantageKit in January.
ENDGAME At Monterrey an elevator formed part of the main structure. Before Hermosillo the team removed it. The team states it almost never climbed in quals and that climbing was not needed in playoffs. After Hermosillo it planned a climber with an improved structure; the final climb is unconfirmed.	**AUTONOMOUS** No working autos at Monterrey, per the team. For Hermosillo the team wrote a 3 piece auto and an amp auto that places one or two notes in the amp. Paths use PathPlanner; a note-align command runs at the end of each path. The team planned a move to Choreo after Hermosillo.
VISION Limelight 3 running a team-trained neural network for note detection, trained on over 53,000 pictures. Used mainly in auto to align the robot to a note and start the intake. Localization ran on LimelightOS; after Hermosillo the team planned a move to PhotonVision and more than one odometry camera.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A mid-season simplification with stated reasons: the elevator broke easily, carried the shooter, and served a climb the team almost never used in quals.
- A short auto can set match strategy: the team used an amp auto to amplify early and reports a +20 point lead in its playoff matches.
- Note detection by motor current was unstable on the field for this team; it changed to a 50cm beam break at its first event.
- A robot-wide SuperStructure state machine, inspired by 254 and 1678, conflicted with the mechanisms at Monterrey. The rebuild added end conditions to all commands.
- The thread gives a full 2024 SysId routine for swerve, logged with the WPILib callback to a FAT32 USB drive on the roboRIO.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 build thread (CD) · Code (GitHub) · Note-align command (GitHub)

■ Study pick

2491 NoMythic Eurydice

USA-MN

Local EPA 27 (#475)

17-8

'Eurydice': 26x26 in swerve; bottom-pivot two-roller shooter that also scores amp; under-bumper intake; telescoping climb

SHOOTER (SPEAKER) Bottom-pivot shooter with a 'sandwich' of two rollers, one on top and one on the bottom (January 19 concept post). The same shooter scores the amp. At Northern Lights the amp shot needed the robot an inch or two back from the amp, good side-to-side alignment and an extreme shooter angle. For 10,000 Lakes the team planned a button that backs the robot up, angles the shooter and scores on its own. Motors and ratios unconfirmed (see CAD).	**INTAKE** Under-the-bumper ground intake with two horizontal rollers. Prototypes showed it can take notes from the front and back, and compliant rollers intook faster than rigid ones. Changed in March after Northern Lights: bottom roller gearing went from 3 times to 2 times drivetrain speed for more force and fewer stalls. Alignment bars pushed notes away, so the team replaced them with ramps under the drivetrain.
FEEDER / AMP / TRAP Indexer design was left open in the January concept. Notes sometimes jammed at the hand-off from intake to indexer, so in March the team added powered side wheels to push the note through. Amp scoring uses the shooter. Trap mechanism unconfirmed (not stated in the public release).	**DRIVETRAIN** 26 in x 26 in swerve, per the January 19 concept post. The team anodized its swerve modules purple in the 2023 offseason. Module type unconfirmed (see CAD). The team's V3 wheel guards came through Northern Lights with one small dent.
ENDGAME Two telescoping tube climber arms, one on each side, with self-centering hooks that align on the chain (January concept). At Northern Lights the climbers worked but the climb was slow, and the 1/4 in aluminum hooks bent side to side. The team prepared spare hooks. Trap and harmony unconfirmed (not stated in the public release).	**AUTONOMOUS** The team set a goal of a four-note auto for 10,000 Lakes: one preload plus three collected notes, and several autos to fit different alliance partners. Whether these ran at the event is unconfirmed (not stated in the public release).
VISION A Limelight was used for object detection on the 2024 robot, per the team's 2025 build thread. Odometry-based shooting worked the night before Northern Lights and failed at the event. The team then added set shots: the driver goes to a known point such as the podium and the robot shoots with hard-coded, pre-tested yaw and pitch. The team states the drivers rarely used this option, so long shots were taken only when needed. Goals for 10,000 Lakes included automatic speaker aim and shooting while moving.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Intake roller speed is a trade against force: the team geared its bottom rollers down from 3 times to 2 times drivetrain speed so they stall on notes less often.
- Plan a fallback for vision: when odometry shooting failed at the event, the team added shots from known points with pre-tested yaw and pitch. The drivers had little practice with them and rarely used them.
- Support the end of a small motor pinion: the team sheared NEO 550 shafts and added a bearing at the end of the pinion to stop the cantilever load.
- Amp shots from a pivot shooter are sensitive to distance; the team planned a button that backs the robot off the amp a fixed amount before scoring.
- Check climber hooks for side loads: the 1/4 in aluminum hooks bent side to side at the first event.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 Open Alliance build thread (CD) · Reveal 'Eurydice' (CD) · Reveal video

■ Study pick

7525 Pioneers

USA-TN

Local EPA 27 (#523)

22-13-1

Static-shooter swerve robot (SDS MK4i); beater-bar intake; roller amp mechanism added for Worlds; two rope climbers

SHOOTER (SPEAKER) Static (fixed) shooter, per the team's amp mechanism post. At Arkansas notes sometimes caught on a small lip when shooting; the team trimmed the lip further and reports no more stuck notes. Shooter motors and ratios are unconfirmed (see CAD).	**INTAKE** Roller intake with a beater bar in front to take hits and protect the rollers. The robot senses a note by motor current. At Arkansas the bar bent and rubbed the rollers, and tape in the gears stopped intaking in the last playoff match. At Buckeye a hit bent the 3/4 in beater tube; the team changed it to 1/2 in hex shaft and added an operator button to turn off the current sensing.
FEEDER / AMP / TRAP Amp mechanism. At Arkansas and Buckeye it was a round tube that flipped up to keep the note from bouncing out of the amp; the team stopped amp shots at Arkansas. Changed for Worlds to a 118-inspired design: a hex shaft with 2 in flex wheels and 3 polycarbonate rollers that bend the note down into the amp. The pivot uses 2 NEOs on a 56:1 reduction and the rollers use a Kraken at 2:1. The team reports 90-95% consistency in testing. Trap is unconfirmed.	**DRIVETRAIN** Swerve on SDS MK4i modules with SPARK MAX controllers near the modules. In October 2023 the team assembled a new set of MK4i modules and described them as NEO modules. Changed before Worlds to 4 Kraken X60 drive motors and the SDS L3+ upgrade kit; the team reports a speed change from about 14.5 fps to 18 fps on the field.
ENDGAME Two climber arms that use Dyneema rope, zeroed by current sensing. At Arkansas one rope snapped in the first playoff match, the third snap on the same side; the team suspects an abrasive point inside the climber. The left arm's current-sensed zeroing also failed at random points. Trap and harmony are unconfirmed.	**AUTONOMOUS** Unconfirmed routines. The team worked with PathPlanner in preseason and won the Autonomous Award at the Arkansas Regional. Note counts for the competition autos are not stated in the public release.
VISION AprilTag vision. In a 2024-09-22 recap the team states the 2024 robot had only AprilTag vision, with no note detection, and drove to pose for automatic scoring. In preseason the controls team worked on Limelight AprilTag tracking for aligning in auto and while scoring. Camera model on the competition robot unconfirmed. After Arkansas the team listed AprilTag alignment for the amp as a planned fix.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Log each event failure with a short-term and a long-term fix. The team's Arkansas and Buckeye posts list rope snaps, sheared bumper bolts, a bent beater bar and a breaker hit by loose wires.
- An amp mechanism must suit the shooter that feeds it. The team found a flip-up tube inconsistent with its static shooter and changed to a roller mechanism with stated motors and ratios.
- Sequence in code can fix a mechanism. The team took the note into the amp mechanism first, then lowered the angle, then scored, and added side standoffs for alignment.
- Give the driver an override for any sensor that can fail. After the beater bar bent and set off the intake current sensing, the team added a button to turn the sensing off.
- A mid-season drive upgrade can be scoped and measured. The team changed to Kraken X60 drive motors and the SDS L3+ kit and reports about 14.5 fps to 18 fps.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 Open Alliance build thread (CD)

Profile	1038 Lakota Robotics Vio-Linda USA-OH	Local EPA 25 (#663) 14-16

'Vio-Linda': REV MAXSwerve; under-bumper intake; 'TRAG' amp/trap scorer that feeds flywheel launchers; chain climb

SHOOTER (SPEAKER) Flywheel launchers for the speaker. The 'TRAG' rollers reverse direction to feed the note into them. Motors and ratios unconfirmed (see CAD).	**INTAKE** Under-the-bumper intake. A 3D-printed transition mechanism moves the note from the intake to the TRAG.
FEEDER / AMP / TRAP The 'TRAG' scores in the amp and the trap. It rides on rack-and-pinion-like vertical slides with metal sprockets. Side pulleys act as rollers to orient the note for drop-off.	**DRIVETRAIN** REV MAXSwerve modules with a frame of REV ION channels, the same base approach as the team's 2023 robot.
ENDGAME A lifting mechanism climbs onto the stage chain, and the robot scores in the trap. Mechanism detail unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the release post; see code).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- One mechanism serves amp, trap and speaker feed: the TRAG scores the amp and trap, and reverses its rollers to feed the flywheels.
- A 3D-printed transition can bridge an under-the-bumper intake and an upper scorer.
- Reuse a known drive base: the team ran REV MAXSwerve on REV ION channels in 2023 and again in 2024.
- CAD and Java code are both public, so a student can trace each mechanism from CAD to code.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · CAD and code release thread (CD)

Profile	3749 Team Optix USA-CA	Local EPA 25 (#622) 17-9

Single-jointed arm carrying a four-bar intake that also shoots; two Limelights for pose-based aim; SDS MK4 swerve

SHOOTER (SPEAKER) The final design from build week 2 is a single-jointed arm carrying a four-bar intake that also acts as the shooter. The arm angle for each shot comes from the robot's field pose. At the San Diego Regional the team reports two runs of 5 matches with no missed Speaker shot from anywhere closer than the podium. The team gives about 95% accuracy inside a marked zone. Shooter motors unconfirmed (see CAD).	**INTAKE** Four-bar intake with polycarbonate rollers, turned by a REV NEO 1.1 on a SPARK MAX. In build week 1 the intake goal was to score in the Amp and pass notes to the shooter. At the San Diego Regional the intake motor oscillated on practice day and acetal parts gave out; the team fixed it. In Qualification Match 76 the intake NEO 1.1 and its SPARK MAX failed, and the team replaced both.
FEEDER / AMP / TRAP The intake also acts as the shooter, per the build week 2 post, so notes pass from intake to shot in one mechanism. The build week 1 intake goal included Amp scoring. Amp and Trap method in the final robot unconfirmed (see CAD).	**DRIVETRAIN** Swerve with SDS MK4 modules, assembled in build week 1. 3D-printed drivetrain corners, inspired by 3847, extend the frame perimeter by 1/8 in and give fastener tolerance. Bumper mounts use a 1/4-20 bolt that slides into a slotted plate (3847/1678 inspired). The drivetrain uses SPARK MAX controllers; drive motors unconfirmed (see CAD).
ENDGAME Build week 1 climb concepts were an angled elevator and a shooter-elevator combination, with Penn State's Ri3D sliding design as a separate idea. The final climb is unconfirmed (see CAD).	**AUTONOMOUS** A 4 note auto was a priority after the Ventura County Regional and ran at the San Diego Regional. The team says it was often interrupted in qualifications by partners who wanted the same start position. Path tool unconfirmed (see code).
VISION Two Limelights give field-pose estimation, with a Pigeon 2.0 gyro added for the San Diego Regional. The pose sets the arm angle for each shot and auto-aligns the robot to the Speaker.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write failure reports: the team logs match, issue, cause, pre-match factors and solution, and plans to inspect spare motors and controllers before events.
- Check the VRM fuse: the team ran the VRM on a 20A fuse until a rule R617 violation forced a change to 10A during playoffs at the San Diego Regional.
- Aim from pose: two Limelights and a Pigeon 2.0 give the field pose that sets the arm angle and the robot heading for each shot.
- Print drivetrain corners: 3D-printed corners add 1/8 in to the frame perimeter and give tolerance for fasteners.
- Set goals before designs: at kickoff the team writes a need, want, wish list with no designs allowed.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 build thread (CD) · Kickoff need, want, wish list (Google Sheets)

■ Study pick

6838 X-SHARC Blackout

Turkiye

Local EPA 25 (#660)

15-11

'Blackout': swerve; shooter on a pivoting arm; touch-and-grab ground intake; amp shot backward from the feeder

SHOOTER (SPEAKER) The shooter mounts on a pivoting arm; the team tuned arm PID on 2024-02-17. The shooter wobbled in early tests, and the team built a new shooter iteration before the first regional. At Istanbul Regional the step inserts in the tube plug that held the shooter to the arm tore out; brackets replaced them. Flywheel motors and wheels unconfirmed (see CAD).	**INTAKE** Touch-and-grab ground intake, a stated build goal. A 3847-style passive 3D-printed centering part was added on 2024-02-18 and kept. The intake motor changed from a 775-class motor to a NEO after motors burned out while notes stuck at the feeder. The steel intake plates bent repeatedly at Halic Regional; the team made its 8th or 9th plate change in the finals.
FEEDER / AMP / TRAP Feeder: the team tried 2 in 60A with 2.25 in 30A compliant wheels, then a NEO; notes still stuck. The fix was polyethylene film on the shooter bottom plate and a film-covered wedge; team 8159 recommended the film. Tennis racket grip on the front feeder roller helps feed the note. Amp: the hood design failed, so the robot shoots into the amp backward from the feeder with no hood. No trap mechanism is described.	**DRIVETRAIN** Swerve; the March 2024 update reports a battery dip to 7v when the swerve drive motors draw power. Module type unconfirmed (see CAD). The team's 2025 blog says it is 'still using MK4i swerve drive'.
ENDGAME The climb design uses the arm. The team designed a chassis hook that latches on the chain to hold the robot. Its chain tensioner could not hold the climb loads, so the team did not climb at Istanbul Regional. Climb use at Halic Regional unconfirmed.	**AUTONOMOUS** At Istanbul Regional the robot had a 2-note auto and a 4-note auto that collects the three notes in front of the speaker. After the event the team had a 4-note auto with far shots and was developing a 5-note auto. Path tool unconfirmed.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** 'Blackout' CAD (Onshape)

LESSONS FOR STUDENTS

- Check where the intake pushes the note before you change wheels or motors. The team found the note pressed into the shooter bottom plate, and low-friction polyethylene film plus a wedge fixed the transfer.
- Match plate material to impact loads: the steel intake plates needed 8 or 9 replacements at one event, and 3mm polycarbonate plates fixed both the bending and the loose centering pieces.
- Test the climb with full loads before an event; the chain tensioner failed in normal climbs, so the team did not climb at its first regional.
- Fasten the shooter to the arm with parts rated for the loads; tube-plug step inserts tore out at the first regional.
- Comm loss in five qualification matches stopped after the team changed the radio power module to a PoE injector cable. The team says it is not sure this was the cause.

Onshape CAD: 'Blackout' CAD (Onshape) · 'Blackout' CAD (Onshape) · 2024 build blog and CAD release (CD) · 2025 build blog with 2024 offseason recap (CD)

Profile	2423 The KwarQs ConDUCKtor USA-MA	Local EPA 24 (#731) 20-23

'ConDUCKtor': over-the-bumper intake; one pivot for shooter and climber; swerve (team's second swerve year)

SHOOTER (SPEAKER) Pivoting shooter. The same pivot is used to climb, per the release post. The team redesigned the shooter between DCMP and BattleCry at WPI. Flywheel type, motors and ratios: Unconfirmed (see CAD).	**INTAKE** Over-the-bumper intake. The team states it went through several intake iterations during the competition season. Rollers, motors and deploy method: Unconfirmed (see CAD).
FEEDER / AMP / TRAP Unconfirmed (not stated in the public release). The post does not describe a feeder, amp mechanism or trap mechanism.	**DRIVETRAIN** Swerve. The team states 2024 was its second year using swerve drive, run with YAGSL. The release shares bolt-on MAXSwerve covers and 3D-printed TPU MAXSwerve wheels that the team developed in the 2023 offseason; the team states they served it well. Gear ratio: Unconfirmed (see CAD).
ENDGAME Climb with the pivoting shooter/climber, per the release post. Trap and harmony: Unconfirmed (not stated in the public release).	**AUTONOMOUS** Autonomous modes built in PathPlanner. The team also tried PathPlanner pathfinding to drive to scoring positions in teleop. Note count per routine: Unconfirmed (see code).
VISION PhotonVision for AprilTag detection. The team also used the PhotonVision object-detection pipeline to automate picking up game pieces. Camera count and model: Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** 2024 robot CAD (Onshape)

LESSONS FOR STUDENTS

- One pivot serves as both the shooter angle joint and the climber. Open the CAD to see how one mechanism covers two jobs.
- The configurable crush block for 2x1 tube labels each configuration by wall thickness and uses teardrop holes for 3D printing. The team later added a 1x1 version after forum feedback.
- Bolt-on MAXSwerve covers let the team clean and re-grease modules without taking them apart.
- The team moved from mostly custom code to YAGSL and command-based Java and states this got the swerve running early so it could focus on other subsystems.

Onshape CAD: 2024 robot CAD (Onshape) · CAD and code release (CD) · 2024 robot CAD (Onshape) · Code (GitHub) · Configurable crush block (Onshape) · 3D-printed bumper mounts (Onshape) · MAXSwerve screw-on covers (Onshape)

■ Study pick

5987 Galaxia in memory of David Zohar Andromeda

Israel

Local EPA 24 (#693)

30-26

Note robot on flipped swerve; polycarbonate spring-hook Stage climber with a locking cable winch

SHOOTER (SPEAKER) Shooter type unconfirmed (see CAD). At the Israel District Championship the robot focused on the Speaker in a three-offense alliance. Before shipping to the Championship, the team spent most of its repair time on long-range shooting.	**INTAKE** Intake type unconfirmed (see CAD). In a District Championship playoff match the robot lost a belt on its intake and switched to defense.
FEEDER / AMP / TRAP Feeder, Amp and Trap mechanisms unconfirmed (see CAD). At the District Championship the team describes delivering Notes to the Amp for its partners to score.	**DRIVETRAIN** Swerve with flipped modules. A Dec 2024 team post says the team used flipped swerve modules for the past two seasons. Module model unconfirmed (see CAD).
ENDGAME Stage chain climb. Two polycarbonate strips, 8mm thick and 40mm wide, carry hooks at their ends. Cables run from the hooks to a winch gearbox that the team repurposed from its elevator. The cables hold the strips below Stage height; letting out cable lets the polycarbonate spring the hooks up. If the hooks rise too high, the team runs the cable into the Chain to pull them down until they catch. The robot winches in and locks the gearbox to stay up after T=0. Changed after Champs: the team replaced the lock servo with a BAG motor. Trap and harmony unconfirmed.	**AUTONOMOUS** Two 4-Note routines, one on the near side and one on the far side of the Wing, shown in a team post during the district events. At that time the team said the far-side routine still needed work to finish under 15 seconds. Path tool unconfirmed.
VISION Unconfirmed for the competition robot (not stated in the public release). In December 2023 the team's off-season vision setup used Orange Pi 5 coprocessors, Arducam OV2311 cameras and PhotonVision, plus a Limelight. A Dec 2024 team post says inefficient vision code caused loop overruns at an October 2024 off-season event.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A spring hook can be two polycarbonate strips: the team's climber uses the plastic's spring force to raise the hooks when the winch lets out cable.
- Design the hold after T=0 with care: the team says the gearbox lock was its only major mechanical problem at Champs, and it replaced the lock servo with a BAG motor.
- Reuse parts the team already trusts: the climb winch uses a gearbox the team repurposed from its elevator.
- Organize CAD for parallel work: before the season the team moved from SOLIDWORKS to Onshape and used one document per subsystem to limit load times and accidental changes.

Onshape CAD: Onshape CAD · Onshape CAD (Spectrum CAD Collection) · 2024 build thread (CD) · Reveal thread (CD) · Reveal video · Off-season notes on the 2024 robot (CD, 2025 build thread)

■ Study pick

3737 Roto Raptors Synthwave

USA-NC

Local EPA 22 (#850)

25-18-2

'Synthwave': 3D-printed swerve; under-bumper intake; through-robot note path; sprung single-hook climber; no trap

SHOOTER (SPEAKER) The release post names a 'cannon' that rotates with torsion spring assist and raises the climber lift arm. The post does not say the cannon is the shooter; shooter wheel type, motors and ratios unconfirmed (see CAD).	**INTAKE** Under-bumper ground intake. The spring-loaded roller assembly rises for more ground clearance when the robot is not intaking. No mechanisms extend outside the chassis perimeter.
FEEDER / AMP / TRAP Through-robot note handling with dual belts to move the note. The team states the robot could do every game task except the trap. Amp scoring detail unconfirmed (see CAD).	**DRIVETRAIN** Compact swerve modules of the team's own design, printed in PETG on FDM printers. The release names NEO drive and NEO 550 steer motors as the base design, with SPARK MAX controllers; calculated free speed with a NEO is 15.41 ft/sec. Steering pairs the NEO 550 with UP 4:1 and 3:1 gearbox stages and turns on 1/4 in Delrin balls in printed PETG V-grooves. Each module has a low-cost absolute encoder; a 2024-01-29 update says to remove resistor R4 so the encoder works. Wheels are the team's 3D-printed TPU suspension treads on a printed PETG hub. The robot is under 14.5 in tall in starting configuration, which the team states permits maximum speed with no tipover. The team calls it its fastest robot. Changed for 2025 to NEO Vortex drive motors.
ENDGAME Single-hook chain climber. The hook is articulated and sprung to remove moment on the arm while hanging, and its profile resists slipping. The cannon raises the lift arm to reduce motor requirements. Compact needle roller bearings held by 3D prints guide the lift arm extension. Two servo-operated locks prevent a drop after the match. No trap.	**AUTONOMOUS** Unconfirmed (not stated in the release post; see code).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Robot CAD (STEP, Google Drive)

LESSONS FOR STUDENTS

- An FDM 3D-printed swerve module can run a full season: the team reports no module failures over 2024, at about $200 per module without motors and controllers. The release includes a BOM, print settings, assembly steps and a speed calculator.
- Read the encoder notes before building: a 2024-01-29 update says the assembly instructions left out a link wire the encoder needs, and the team's fix is to remove resistor R4.
- Tune a printed tread to robot weight: the team sets TPU infill at 40% for a 90 lb robot and 50% near max weight, and reports a tread life of two competitions.
- Use one mechanism to help another: the cannon raises the climber lift arm, which reduces motor requirements.
- A sprung, articulated hook removes moment on the arm while hanging; two servo locks hold the robot after the match.
- Design for service: two screws remove the bumpers, and clamshell 3D-printed cable management allows quick changes.

Drive folder / doc: Robot CAD (STEP, Google Drive) · Robot CAD (STEP, Google Drive) · Electrical board CAD (STEP, Google Drive) · CAD and code release thread (CD) · 3D-printed swerve release (CD) · 3D-printed suspension treads release (CD) · 3D-printed swerve files (Google Drive) · 3D-printed suspension tread files (Google Drive)

■ Study pick

6925 W.A.Robotics VENDETTA

USA-GA

Local EPA 22 (#878)

26-21

'VENDETTA': fixed-angle 2-Kraken shooter; over-the-bumper intake with handoff; two hook climbers; MK4i swerve

SHOOTER (SPEAKER) Top and bottom shooter rollers, each powered by an independent Kraken. In January the team wrote that the shooter angle should be 60 degrees; it considered an adjustable shooter and chose not to build one. In February tests, 65% speed gave consistent speaker shots up to 14 inches away; higher speeds damaged notes and drew more power. Side shots and lofted long shots were less accurate.	**INTAKE** Over-the-bumper intake with custom plates cut on an X-Carve. The team states it picks up notes that are not centered. A limit switch is mounted on the intake with a custom bracket; its use is not stated. The pivot motor uses an 80:1 MAXPlanetary gearbox, added in February. That gearbox was built wrong and was replaced in late February. In February the team started a second intake and reported it almost finished.
FEEDER / AMP / TRAP The intake hands the note to the shooter; a curved backplate added on 2024-01-31 holds the note for the handoff. Amp: the original plan was to score with the intake, which the team tested. The shooter also scores in the amp: 29 of 29 shots at 35% speed. Trap: Unconfirmed (not stated in the public release).	**DRIVETRAIN** Swerve with SDS MK4i modules. The team planned 8 Krakens and in February waited for the MK4i Kraken adapter kits with 16T drive pinions. The posts do not confirm when the Krakens went on the drivetrain. Swerve code runs on Phoenix 6.
ENDGAME Two climbers with custom hooks. In early tests one climber ran faster than the other at the same speed; the team considered a churro to link them. On 2024-03-13 the team tested the hooks on the chain, including a climb on the side of the chain. Trap and harmony: Unconfirmed.	**AUTONOMOUS** The team tested 4-piece and 5-piece autonomous paths in January 2024, before the competition robot was finished. Final competition routines and path tool: Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** 2024 robot CAD (Onshape)

LESSONS FOR STUDENTS

- The team chose a fixed-angle shooter for a team of about 15 members and 6 veterans. Read the January post for the reasoning.
- The February shooter tests give a method a small team can copy: vary speed and position, record hit rate, and stop tests that fail early.
- Shooter speed is a trade: higher speed improved accuracy but damaged notes and drew more power, so the team set 65%.
- Two scoring paths for the amp, intake and shooter, give a backup if one subsystem breaks. The team tested both.
- One note shot 235 times showed visible wear. Keep new notes for competition tests.

Onshape CAD: 2024 robot CAD (Onshape) · 2024 robot CAD (Onshape) · 2024 Open Alliance build thread (CD) · Code (GitHub) · Reveal 'VENDETTA' (CD) · Reveal video

■ Study pick

4276 Surf City Vikings

USA-CA

Local EPA 19 (#1150)

15-9-1

Intake/shooter on a NEO-driven 4-bar, rebuilt mid-season; floor intake; MaxSwerve; climber on separate motors

SHOOTER (SPEAKER) Combined intake/shooter unit on a 4-bar arm. In January the team tested a left/right wheel shooter and reported it working. The competition shooter is, per the team, 'pretty much identical' to the wood prototype, with belts instead of chain. Shooter motors and wheel sizes unconfirmed (see CAD). The 4-bar first used a modified REV Starter Bot drive tower at about 250:1 on one NEO. Changed after Ventura: doubled arms, 1/4 in mounting plates in place of MaxTube, two 4-bar motors.	**INTAKE** Floor intake with top/bottom wheels in the same unit as the shooter. A NEO 550 on 3:1 drives 2.25 in AndyMark compliant wheels. It stalled on floor pickup, so the team changed to 5:1 in week 6. The first intake was wood; the team replaced it with polycarbonate.
FEEDER / AMP / TRAP An E18-D80NK beam-break sensor detects the note in the intake/shooter. Reflection off polycarbonate and the aluminum brace blocked it; matte black grip tape on the brace fixed it. In a build-season test the team ran the upper roller in reverse to place notes in a mock amp. Trap mechanism unconfirmed.	**DRIVETRAIN** Swerve: the team built a new MaxSwerve chassis. At Ventura the drive motor keys did not stay in place. At Orange County the team reports no swerve drive issues. During build season a bent pin on a through-bore encoder stopped one module's turning reading.
ENDGAME Two climber assemblies. At Ventura the climber was driven from the 4-bar, which the team calls disastrous. Changed for Orange County: the climber got its own motors, and the team reports it worked. A spring-loaded multihook add-on was in work in week 6; its use is unconfirmed.	**AUTONOMOUS** January plan: score in the speaker or amp, then leave. Final auto routines unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release). An auto shot alignment was added in the 2024 offseason, before Beach Blitz; its sensor is not named.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Stress-test the robot before the first event; the team lists skipping this as a cause of its Ventura failures.
- One NEO driving both sides of a 4-bar through a shaft at about 250:1 was wobbly and failed; doubled arms, 1/4 in plates and two motors held up at Orange County.
- Give each mechanism its own drive: the climber driven from the 4-bar failed at Ventura and worked on separate motors at Orange County.
- Check intake gearing under load: a NEO 550 at 3:1 stalled on floor pickup, and 5:1 worked.
- A beam-break sensor can fail on reflective surfaces; matte grip tape on the brace behind it fixed the reading.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 Open Alliance build blog (CD) · 2024 offseason build blog (CD)

■ Study pick

663 Power Sharks Hammerhead

USA-MA

Local EPA 17 (#1418)

18-23

'Hammerhead': Ri3D-based wide intake-shooter; sliding 8020 extending climbers on #35 chain

SHOOTER (SPEAKER) Combined intake-shooter, based on the 2024 Unqualified Quokkas Robot in 3 Days design. The team made it wider than the game piece to make intaking easier, and states this made accurate shooting harder. Motors and ratios unconfirmed (see CAD).	**INTAKE** The intake is the wide intake-shooter. Its large span took hits at competition. The 6000-series churro tubing used to save weight yielded quickly; the team replaced it at the event with 7075 aluminum hex stock, which held up better. The team notes that some other Quokka-style robots put the intake inside the frame.
FEEDER / AMP / TRAP No separate feeder, amp or trap mechanism is described in the release post. Unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The team used 8020 extrusion in many places instead of box tube, with through-bolted connections. The team states weight was not an issue for this design.
ENDGAME Sliding-contact extending climbers on 8020 slides, driven with #35 chain. The team states they support side-by-side buddy climbing on the chain. Through-bolts on the load path are needed because climb loads can make slide-nut connections slip.	**AUTONOMOUS** Unconfirmed (not stated in the public release)
VISION No camera system is stated. The team used Grapple Robotics laser sensors for game piece detection.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Size structural stock for impact, not only weight: 6000-series churro yielded on the wide intake, and 7075 hex stock replaced it at the event.
- Through-bolt the climb load path; the team states slide-nut connections can slip under climb loads.
- A wider intake-shooter trades shot accuracy for easier intaking, by the team's own account.
- A Robot in 3 Days design can be a season base: Hammerhead follows the 2024 Unqualified Quokkas Ri3D robot.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 CAD release thread (CD)

Profile	1072 Harker Robotics USA-CA	Local EPA 16 (#1561) 11-19

Two-stage cascade elevator carrying a pivoting top-down shooter; over-the-bumper ground intake; centering indexer; MK4i swerve

SHOOTER (SPEAKER) Top-down shooter on a pivot mounted on the elevator. Early prototypes used Colson wheels as flywheels. A Falcon-powered prototype reached about 8 feet; adding Cat Tongue grip tape tripled the distance, but the tape flung loose. Final motors and ratios unconfirmed (see CAD).	**INTAKE** Over-the-bumper ground intake. The hand-run prototype used aluminum rollers with silicone tubing. The team found it was nearly impossible to shift notes left and right, so it needed a centralizing mechanism.
FEEDER / AMP / TRAP An indexer between the intake and the shooter centers the note and improves the handoff. The two-stage cascade elevator raises the pivoting shooter; the team chose this layout to score in the Trap. Amp method unconfirmed (see CAD).	**DRIVETRAIN** Swerve on SDS MK4i modules. The team switched from MK4 to MK4i for 2024, stating MK4i is safer for the Falcons. Modules bought before June used #8-32 mounting while later parts used #10-32, so mixing parts caused hole mismatches. The robot uses two CANivores, one per drivetrain half, because devices on longer stub lengths did not register.
ENDGAME The team's priority list includes a 2 Onstage climb and the Trap. Climb mechanism unconfirmed (see CAD).	**AUTONOMOUS** In February 2024 the team planned a six note autonomous path. Final routines unconfirmed (see code).
VISION Limelight AprilTag pose estimation. The team loaded the field map into the Limelight pipeline and read robot poses from NetworkTables.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Let one scoring goal drive the architecture: the team chose an elevator-mounted pivoting shooter because it wanted to score in the Trap, and published its architecture comparison.
- Add a centering stage when the intake cannot center the game piece: the team found it nearly impossible to shift notes sideways in the prototype, so an indexer centers the note before the handoff.
- Check part revisions before mixing hardware: the team found MK4i parts ordered before June used #8-32 mounting while later parts used #10-32.
- Split a CAN FD bus when devices on long stubs do not register: the team used two CANivores, one per drivetrain half.

Onshape CAD: Onshape CAD · Onshape CAD · 2024 build thread (CD) · Code (GitHub)

■ Study pick

1746 Team OTTO

USA-GA

Local EPA 16 (#1591)

30-17

CTRE swerve; Falcon shooter fed by a primer; intake rebuilt on a Falcon; PathPlanner and Choreo autos

SHOOTER (SPEAKER) Shooter rollers on Falcon motors. The team states SysID worked for the shooter: they entered the Falcon constants and it worked at once. Wheel type and ratios unconfirmed (see CAD).	**INTAKE** Intake v1 was ineffective; drivers often had to push notes against a wall to intake. It burned 3 NEO motors and their SPARK MAX controllers with current limits applied. A Falcon replaced them and worked.
FEEDER / AMP / TRAP A primer feeds the shooter. It ran on a 775 motor, which did not have enough torque or speed to shoot the note consistently. Primer feed was inconsistent and sometimes hit the rollers, so the team could not reliably run the intake sequence with the shooter rollers running. At the District Championship the team stored notes in the intake while ferrying as a workaround. Amp and trap detail unconfirmed (see CAD).	**DRIVETRAIN** Swerve, run with CTRE swerve code that the team states integrated without problems. Module type unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** Paths built with PathPlanner and Choreo; the team states both integrated well. Note counts unconfirmed (see code).
VISION PhotonVision, used in simulation during development per the code reflection. Camera count and placement unconfirmed (see CAD).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Size motors to the load: a 775 on the primer could not shoot consistently, and the NEO intake burned out three times before a Falcon replaced it.
- Test the intake early against open-floor notes: v1 needed notes pushed against a wall to pick them up.
- Name commands by how they end: the team found commands with similar names that ended at once, ran forever, or ran until a condition.
- SysID characterization of the shooter worked on the first try with the Falcon constants, per the team.
- Quick merges at events led to redundant code and bloated subsystems, per the team's reflection.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · Code and CAD release with reflection (CD) · DCMP recap video

Profile	9047 SESI SC TECHMAKER Seu Wilson Brazil	Local EPA 15 (#1620) 7-3-1

Low robot: fixed under-bumper intake-shooter; elevator that climbs and carries an Everybot-style amp claw; MK4i L3 swerve

SHOOTER (SPEAKER) Fixed intake-shooter inspired by the RI3D Quokkas robot. The design goal was to score in the Speaker from the Stage, and the team reports consistent Speaker shots with the prototype. In early February the team reported a weakness in the shooter and said it was making it stronger. Motors and ratios unconfirmed (see CAD).	**INTAKE** Wide 'touch it, own it' ground intake below the bumper, combined with the fixed shooter. The team planned a sensor or switch to detect when a note is inside.
FEEDER / AMP / TRAP A claw on the elevator, based on the Everybot 118 claw, picks up from the Source and delivers to the Amp. Trap unconfirmed (see CAD).	**DRIVETRAIN** Swerve on MK4i L3 modules with NEOs, driven in code with YAGSL. Drivetrain 28 in, initial robot height 26 in, low enough to pass under the chain. The design target was a weight below 40 kg.
ENDGAME The elevator is also the climber, with a climb height of 34.6 in. It uses the Compact Elevator Bearing Kit, HTD 5m belts, ABS printed pulleys and a MAXPlanetary gearbox. In early climb tests the belt shifted in its clip, and the team planned a printed HTD 5M clamping plate. The must-have list included hanging on the chain with a partner.	**AUTONOMOUS** Build-season goal of 3 notes, 4 if possible, plus a test route to collect 2 notes from the center of the field. Final routines unconfirmed (see code).
VISION Build-season plan: use AprilTags to navigate to field points, mainly the Source, with a Raspberry Pi co-processor. The team also planned note and robot detection to avoid collisions in auto. Final setup unconfirmed.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write the must-have list before CAD: MK4i L3 under 40 kg, floor pickup, a wide under-bumper intake, Speaker shots from the Stage, Amp scoring, and a chain hang with a partner.
- One elevator, two jobs: it climbs and carries the amp claw, and the robot stays low enough to pass under the chain.
- Belt-driven elevators need a secure belt clamp under climb load: the belt shifted in its clip in early climb tests.
- A week-by-week build thread shows how a second-year team scoped a robot for a week 1 regional.

Onshape CAD: Onshape CAD · Open Alliance build thread (CD) · Onshape CAD · Code (GitHub)

■ Study pick

3596 Rocket Robotics Kerchunk

USA-WI

Local EPA 11 (#2329)

3-7

'Kerchunk': Kitbot launcher on a 27x27 in swerve chassis; under-bumper ground intake; AndyMark Climber in a Box

SHOOTER (SPEAKER) Based on the Kitbot launcher. The build thread states NEO motors on the Kitbot shooter during testing; motors on the final robot unconfirmed (see CAD). In Kitbot testing the team saw a left hook and filmed it in slow motion. The note decompressed while still touching the right shooter rail, and the team planned to change where that rail ends. In the off-season the team added a second set of shooter wheels, which it says improved performance.	**INTAKE** Under-the-bumper ground intake, per the release post. Rollers and motors: Unconfirmed (see CAD).
FEEDER / AMP / TRAP Unconfirmed (see CAD). Amp and trap mechanisms are not described in the release.	**DRIVETRAIN** 27x27 in swerve chassis, per the release post. The team states the 27x27 size fits through its doors with standard bumpers. Module type: Unconfirmed (see CAD).
ENDGAME AndyMark Climber in a Box with modified hooks. The build thread states the students chose two of the 2-stage AndyMark Climbers in a Box.	**AUTONOMOUS** The team used YAGSL and PathPlanner for the first time. Routines: Unconfirmed (see code).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Start from a known base: a design team of three middle school students kept the Kitbot launcher and changed the chassis, intake and climber around it.
- Film the failure: slow-motion video showed the note decompressing against the right rail, which the team took as the likely cause of the Kitbot left hook.
- COTS climbers can carry an endgame: two 2-stage AndyMark Climbers in a Box, with modified hooks, gave the climb.
- Iterate after the season: an off-season second set of shooter wheels improved performance, per the team.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · 2024 Open Alliance build thread (CD)

Sources: the Spectrum 3847 CAD Collection, team CAD, binder and code releases, Chief Delphi release and reveal threads, and match data from The Blue Alliance. EPA is Local EPA (approx), computed independently. It is not official Statbotics data. "Unconfirmed" means the public material does not state the detail.